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SURVEY

A Review of the Current Usage of AI/ML for Radio Access Network (RAN)

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ABSTRACT Future cellular communication is envisioned to rely on intelligent networks enabled by Artificial Intelligence (AI)/Machine Learning (ML) methods. However, research on AI/ML-based networks is still in its infancy, and existing works do not address the whole process of integrating AI technologies into 5G and beyond networks. This systematic review assesses the current state of AI and ML applications for future wireless networks, focusing on four main topics: task offloading, resource allocation, spectrum management, and channel estimation. This work confirms that research in this area faces numerous challenges, including the lack of reliable datasets, limited information on tools and libraries, hardware impact, and simulation platforms. These factors make it difficult to compare and validate the performance of different AI/ML approaches, impairing their progress into standardized solutions. Additionally, the cost of deploying AI is often overlooked, undermining the credibility of the reported gains. Moreover, current surveys rarely analyze the types of datasets used, the runtime cost of models, or the simulation and ML tools adopted. These omissions hinder reproducibility and obscure the trade-offs between performance and implementation cost. The outcome of this review emphasizes the need for more efforts to address these challenges and advance the field of AI/ML-based network research.

INDEX TERMS 6G, RAN, AI, network optimization, datasets.

I. INTRODUCTION

The Fifth-Generation Network (5G), an enabler of Internet of Things (IoT) and Internet of Everything (IoE), follows a communication-centric design principle to empower vertical transformation of application scenarios along with the transition towards cloud-native architecture [1].

The evolution of 5G systems brings ever-increasing complexity and imposes multiple demands on mobile networks [2]. Nevertheless, 5G is still unable to provide permanent solutions to address the increased key challenges posed by a massive volume of data traffic, energy consumption, and the growing social demands [3]. These societal demands encompass a broad spectrum of novel use cases that exceed the capabilities of 5G, including those requiring ultra-low latency on the order of microseconds, e.g., holographic communications [4]. Moreover, it is foreseen that classical computational methods are unable to meet all these requirements, in addition to the need for more modern communication technologies [5]. The challenges highlighted

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by 5G draw attention to AI and ML as a perfect match for 5G, where AI/ML complements 5G operational postures by learning from complex system patterns to provide scope for autonomous operation, transforming 5G into a scalable, real-time network that is data-driven.

The importance of AI was emphasized by standardization bodies, i.e., 3rd Generation Partnership Project (3GPP) Rel-17, which investigates the integration of AI applications into the Radio Access Network (RAN). This is further emphasized by the ramping up of AI items for the 5G evolution [6]. The ongoing 3GPP Rel-18 will mainly focus on 5G-Advanced with evolutionary updates, where the integration of AI/ML will trigger a paradigm shift in future wireless networks, providing enhancements for the physical layer and RAN [7].

The applications of AI/ML in 5G networks include network planning, automating network operations, e.g., optimization and security, and enhancing both the quality of service and the customer experience. Moreover, AI technology is being deployed to tune performance across all layers, starting with the distributed cloud layer (5G Edge/Core), moving to Integrated Access Backhaul (IAB), and finally to disaggregated RAN.

According to Ericsson's 2024 report on AI in RAN [8], AI is becoming integral to RAN operations, transforming how networks manage complexity, optimize performance, and improve Energy Efficiency (EE). Ericsson highlights that AI-powered features, including link adaptation, MIMO sleep mode, and cell shaping, have led to measurable improvements, such as up to 14% energy savings per site and a 50% increase in downlink throughput at the cell edge. Moreover, AI-enabled mobility enhancements have reduced inter-frequency handover failures by 10.9% and overall call drop rates by 1.2%. The report emphasizes that AI's role is expanding from slow control loop optimization to fast loop decision-making, paving the way for AI-native, intent-driven RAN architectures that support real-time adaptation and scalable automation.

Accordingly, it has become necessary to replace traditional mathematical models with complex data-driven ML techniques [4], [9]. On the other hand, numerous researchers are interested in integrating AI into wireless communication systems to achieve a significant increase in efficiency, particularly in the domains of wireless resource allocation and management [10], [11]. Utilizing AI will continue to enhance and optimize network autonomous management and control learning systems by incorporating essential technologies, including intelligent core and edge networks. Additionally, Deep Learning (DL) is also expected for channel coding and decoding technology, signal estimation and detection technology, and wireless resource allocation technology [12].

Due to AI's ability to process large-scale data and perform efficient decisions, as well as DL's ease of implementation, AI has proven to be one of the promising technologies that will support Sixth-Generation Network (6G) [13]. AI will support the transition towards cloud-native and AI-native 6G systems [14]. 6G deployment will depend on compute-centric principles [15] and is believed to pave the way for the 'Internet of Intelligence' with connected people, things, and intelligence [16]. Alongside these characteristics, 6G architecture is expected to include highly dense networks with arrays of antennas and with massive amounts of data, where each network node will perform decisions based on that data [17]. On the evolution to 6G, AI will enable the connectivity of billions of heterogeneous devices.

AI will efficiently accomplish seamless connectivity between ground, satellite, and airborne equipment through network data, business data, user data, and other multi-dimensional data-aware learning and real-time, high-speed exchange [18].

Despite growing interest in applying AI/ML to communication systems, existing surveys often overlook critical implementation aspects, specifically the details of AI algorithms, dataset origin, simulation tools, runtime cost, and hardware specifications. These missing details limit the reproducibility and scalability of proposed solutions. This work addresses these shortcomings by providing a structured analysis of such overlooked elements, as discussed in detail in II-C and III-A.

A. MOTIVATION

Despite the growing enthusiasm for integrating AI/ML into 5G and emerging 6G networks, the current body of research lacks coherence and systematization. Most existing works address isolated problems, such as latency reduction or energy optimization, without providing a unified understanding of how AI technologies are being deployed across the broader RAN landscape. This fragmented view hinders the ability to assess progress, replicate methodologies, and identify effective solutions that can transition toward standardization and deployment. Moreover, the diversity of algorithms, tools, training platforms, and evaluation criteria further complicates comparison and validation. Therefore, a systematic review is essential to map the current research terrain, expose critical gaps, and highlight methodological patterns. By classifying the literature according to key AI application areas within the RAN, this work aims to provide a structured foundation that supports scientific transparency and steers future efforts toward scalable, AI-native network architectures.

This work divides AI applications in 5G and beyond networks based on data-driven approaches into four categories: task scheduling and offloading, resource allocation, spectrum management, and channel estimation. After classifying the research articles into these groups, we thoroughly examined each one in light of the following criteria: the objective under each main category in 5G RAN application area, the scenario considered, AI technique/algorithm, the tool/programming language used in both testing and scenario realization, and the type of datasets. The most common AI applications addressed in the reviewed articles are resource optimization, latency minimization, EE, connection optimization, channel

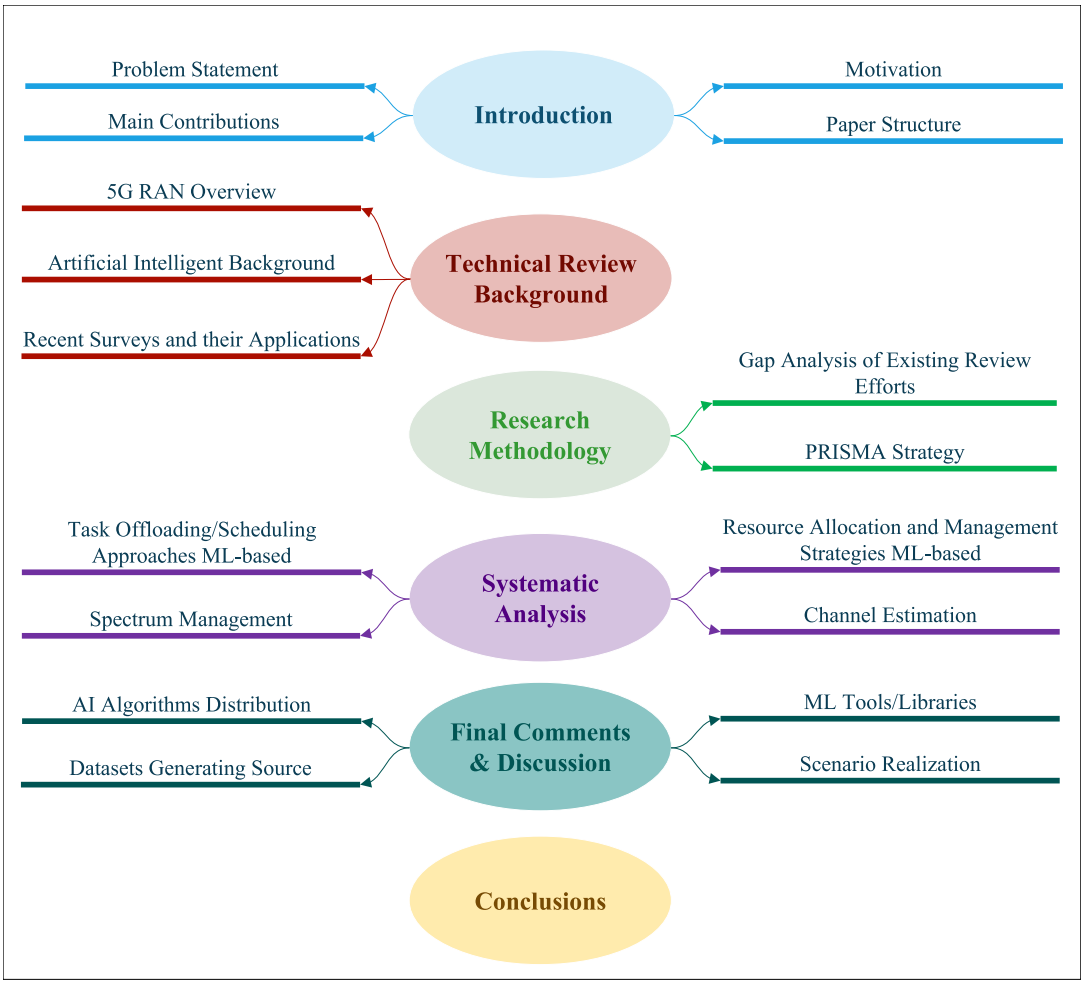


FIGURE 1. Organization of the review paper.

estimation enhancement, spectral efficiency improvement, and radio resource selection and allocation. Most of the revised articles employed one or more AI techniques/ algorithms in their proposed solutions.

B. PROBLEM STATEMENT

This paper presents survey outcomes on current efforts in AI/ ML applications within the RAN area. The systematic review aims to answer the following research questions:

- In what area of the network and in which scenario was ML applied?
- What is the objective to be achieved in that area, besides the importance of the proposed solution?
- What is the methodology used to train the chosen ML algorithm?
- Does the article analyze the class and complexity of the selected ML algorithm?
- Was the used dataset extracted from real data or generated by a simulation tool? What is the name of the simulator used?

All these inquiries will be answered in detail in the following sections.

C. MAIN CONTRIBUTIONS

The contributions of this systematic review are:

- Identifying the current efforts that use AI/ML applications in the RAN area of 5G and future networks.
- Presenting a comparative analysis of the research fields based on data-driven factors, AI techniques, training methodology, the applied scenario to realize the experiment, the used datasets, and the desired objective.
- Discussing future research aspects in the RAN field, highlighting the challenges facing such direction, and pinpointing the missing details in current research efforts in the literature.

D. PAPER STRUCTURE

After this introductory section, the overall paper structure is outlined as follows. A brief overview of RAN in telecommunications technology and the relevant AI background is presented in Section II, outlining the key points considered in the systematic review. Section III reviews the related works that analyze the role of AI/ML in 5G and future networks, and also discusses the research methodology, including the

inclusion and exclusion criteria, search strategy, and data collection. Section IV provides a detailed analysis of how AI is utilized in 5G networks, focusing on key areas like task offloading and resource allocation. Section V offers final comments and discusses remarks under the context of AI for RAN in 5G and beyond networks. Finally, Section VI concludes the work and provides avenues for future research in the domain of AI in next-generation networks. Figure 1 highlights the survey road map.

II. TECHNICAL REVIEW BACKGROUND

This section outlines the key technologies and AI techniques relevant to enhancing 5G and beyond networks, focusing on the role of AI in improving RAN performance and addressing technical challenges.

A. 5G RAN OVERVIEW

The RAN is the part of a telecommunications system that links individual devices to other parts of a network through radio connections. It does this by sending information via radio waves from end-user devices to a RAN transceiver, and then to the main network, which is connected to the global Internet. Thus, it is the component (part) of the network that provides the user's equipment connection to the core network. RAN components include base stations and antennas that cover a specific region depending on their capacity. Its goal is to provide access and ensure resource management on radio sites. RAN is a major component of wireless telecommunications and has evolved through the generations of mobile networks, culminating in 5G.

RAN has long been a fundamental component of telecommunication systems, acting as the bridge between user devices and the core network. In the context of 5G, the RAN has evolved to incorporate highly complex architectures and air interface innovations, enabling ultra-reliable and low-latency communication. The integration of AI and ML into 5G RAN is pivotal for realizing the vision of fully automated, efficient, and adaptive networks. AI/ML-driven intelligence enhances the ability of service providers to manage increasing traffic loads, dynamic user demands, and diverse service requirements with agility and cost-efficiency.

Among the key AI-driven functionalities in the RAN domain, network optimization has emerged as the principal area where machine learning-based automation delivers the highest impact. According to Ericsson [19], network optimization encompasses autonomous functions designed to improve Quality of Service (QoS), user experience, and resource efficiency. It includes technologies like radio frequency shaping, energy-efficient resource management, traffic handling, and mobility prediction. For instance, research in intelligent RAN power management has demonstrated a reduction of up to 31.08% in power consumption [20], showcasing the tangible benefits of integrating AI into RAN operations.

In line with this, our review categorizes the surveyed literature into four core domains, all of which fall under the umbrella of network optimization:

- Task Offloading and Scheduling Approaches (ML-Based): Target the delegation of computational tasks to external nodes (fog, edge, cloud) to reduce device load and latency.
- Resource Allocation and Management Strategies: Focus on the intelligent distribution of wireless resources based on dynamic traffic and user requirements.
- Spectrum Management: Involve AI techniques for efficient spectrum access, interference mitigation, and adaptive utilization in heterogeneous networks.
- Channel Estimation: Leverage data-driven models to improve estimation of channel state information, enhancing throughput and link reliability.

These four categories were selected because they represent the most active and impactful areas of research in applying AI/ML to RAN, as revealed through our exploratory literature review and confirmed by the PRISMA-based systematic selection process. These functions have played a crucial role in telecommunications system design since 2G [21], [22], but with the advent of 5G, their complexity and performance expectations have significantly increased, creating a great opportunity for AI/ML utilization to face such challenges. The AI-enhanced optimization of these domains enables predictive control, real-time adaptation, and cost-effective scalability.

B. ARTIFICIAL INTELLIGENCE BACKGROUND

The field of AI research and application has exploded over the last decade [23]. The variety of models available, along with their excellent results, has led to their presence in our daily lives [24]. In this scenario, the telecommunication industry has undoubtedly benefited greatly, with AI being a cornerstone in 5G and Beyond networks [25].

This section aims to present the main subfields of AI that, according to our review, have made the most significant contributions in this area. First, we will start with ML, a subfield that, due to the ability of its algorithms to learn from data, has seen an unprecedented degree of adoption in the current development of future 5G networks [26], [27], [28]. Then, the Optimization subfield is presented, whose methods have also been widely adopted due to their efficiency in finding solutions [29], [30], [31]. The relationship among the most representative subfields presented in this section is shown in Figure 2.

1) MACHINE LEARNING

From this point forward, we will use the term ''network'' to refer to communication networks, and we will use ''neural network'' to refer to aspects associated with the AI model.

ML is a subset of AI that studies computational algorithms that allow machines to learn by themselves and improve from data analysis without explicit programming [32]. These algorithms intercept statistical, probabilistic, computer science, and algorithmic aspects that allow them to identify patterns in observed data and build models that infer never-before-seen results without having explicit pre-programmed

rules and models [33]. Unlike other types of methods, ML algorithms are not explicitly programmed to solve a specific problem. Instead, they only require a certain number of examples, and their goal is to fit a function that solves the problem at hand. Therefore, the advantage of these methods is that the same ML method can be applied to reasonably different problems without substantially altering the algorithm. The only thing that requires change is the training/learning examples it experiences.

Shallow learning refers to the majority of ML methods proposed before 2006, including Artificial Neural Networks (ANNs) with a single hidden layer [34]. Some examples of such models are Support Vector Machine (SVM), Decision Trees, K-Means, k-Nearest Neighbors (k-NN), and Q-Learning [35]. According to the type of task, they are usually grouped into:

- Supervised Learning (SL): The knowledge is obtained from data sets with examples represented with some type of labeling in the form of input-output. By identifying patterns in the data, the algorithm seeks to derive a function that describes the dependency between the input and output variables.
- Unsupervised Learning (UL): These algorithms do not require that the data in the training set be previously labeled. In this case, the algorithm identifies patterns in the training data to find new properties and relationships between data points. [36].
- Reinforcement Learning (RL): RL algorithms are the best option when an initial training set is unavailable. The learning process is based on real-time trial-and-error interactions with the environment. Given a state, the agent performs an action, receives a numerical reward, and transitions to a new state. This reward is used for optimization [37], and the agent gradually adjusts its parameters to obtain the best long-term reward possible.

An alternative to shallow learning is DL. This one comprises ANNs with more than one hidden layer, called Deep Neural Networks (DNNs), allowing them to represent much more complex relationships between data. However, the inclusion of more hidden layers also leads to the model having more parameters to train and requiring increasingly more computational resources [38]. One aspect in favor of DNNs is that it allows researchers to leverage recent advances in parallel computing, particularly the use of Graphics Processing Units (GPUs) or Tensor Processing Units (TPUs) to perform calculations [39], making them realistic solutions for many problems. Most DNN architectures can be applied in supervised, unsupervised, and reinforcement learning scenarios, as long as some of their hyperparameters are adjusted.

The main component of neural networks is the neuron, and neural networks are often classified according to the type of neuron that is used most frequently within them. In some cases, neural networks are also classified according to the connections between neurons, the training of the model, and the general tasks they solve. Following these criteria, the most frequent types are:

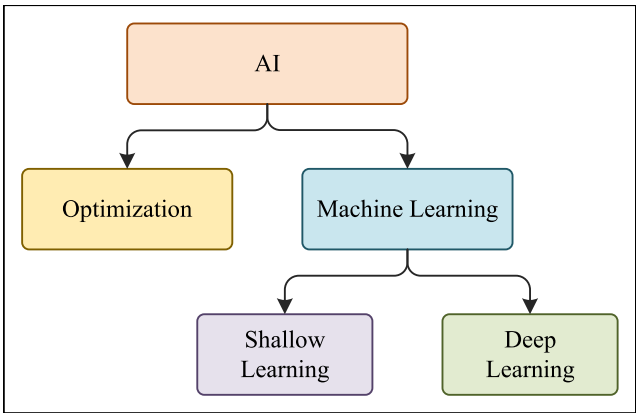


FIGURE 2. AI algorithms main fields.

- Multilayer Perceptron (MLP): It is a feed-forward DNN consisting of an input layer, multiple hidden layers, and an output layer [40]. They are fully connected networks, which means that every node in a given layer is connected to every node in the previous layer. The MLP is the simplest deep learning model in terms of implementation, but it quickly becomes very resource-intensive as the number of weighted connections increases rapidly [41].
- Recurrent Neural Network (RNN): Since MLP cannot benefit from sequential information since each example is computed independently, new types of neurons were created that learn from sequential or time-series data by feeding the output of the previous step as input to the current step. The recurrent [42] neuron was the first proposed neuron, followed by the Long Short Term Memory (LSTM) [43] and Gated Recurrent Unit (GRU) [44] neurons, as the former suffered from a gradient vanishing/explosion when the time dependencies were longer [45].
- Convolutional Neural Network (CNN): If we use an MLP to process an image, the dimension of the input data is usually too high. For example, a simple image of 100×100 creates an input vector with 10,000 features. CNNs solve this problem by processing the 100×100 image directly, rather than a feature vector of 10,000, using a kernel with a much smaller size [46]. Consequently, the number of calculated parameters and the weights required for each layer are significantly reduced, allowing for deeper networks to be built and making them well known for their high predictive power in classification problems involving very high-dimensional data, such as image recognition.
- Generative Adversarial Network (GAN): It is a type of neural network architecture for generative modeling to create new plausible samples on demand. It consists of two competing networks, where the loss of one network is the gain of the other [47].

- Auto-Encoder (AE): It is an ANN trained in an unsupervised way, whose objective is to learn encoded representations of the data, to be reused as input. This allows to learn a reduced information representation, making it a perfect method for compressing information [48].
- Deep Reinforcement Learning (DRL): In shallow RL, a matrix is used to represent the benefit of performing each action given any state, but this becomes too large when the scenario is complex, containing both continuous and discrete variables, with many states and actions. To solve the problem, DRL replaces this matrix with a DNN [49]. The use of the network helps the model when the action-state space is too big to be extensively described in a table, as the network approximates the table with minimum loss.

2) OPTIMIZATION

Optimization methods aim to determine the values of one or more variables that maximize or minimize an objective function, subject to a set of constraints [50]. From the AI perspective, the proposed optimization methods aim to circumvent the significant execution times associated with traditional optimization methods when dealing with complex problems or large volumes of data [51]. As a consequence, intelligent methods are primarily based on heuristics, which, through an intuitive approach grounded in the intrinsic nature of the problem, facilitate solution discovery. Unlike exact methods that provide an optimal solution to the problem, heuristic methods are limited to finding a reasonably good, though not necessarily optimal, solution. Intelligent optimization methods have not only demonstrated their ability to solve a wide variety of challenging optimization problems but have also been used in ML for hyperparameter search, neural network tuning, etc. [52].

C. RECENT SURVEYS AND THEIR APPLICATIONS

AI and ML techniques have been widely applied to improve various aspects of network performance in 5G and beyond networks. In this section, we systematically assess relevant works, categorized into four main areas: Task Scheduling and Offloading, Resource Allocation and Management, Spectrum Management, and Channel Estimation. Table 1 summarizes the extent to which these topics are addressed in the surveys and literature reviews.

1) TASK SCHEDULING AND OFFLOADING

AI/ML can play a crucial role in task scheduling and offloading, where dynamic decisions are made to distribute computational tasks among cloud, edge, and fog computing nodes based on the current network state.

In [53], the authors discuss how AI/ML models predict traffic demand to enable proactive load balancing and mobile edge caching, optimizing task scheduling decisions. Accurate traffic prediction leads to efficient task offloading and reduces network congestion. However, this survey lacks a structured evaluation of dataset sources and AI toolchains.

In another survey [54], the authors highlight how AI/ML techniques are used to improve mobility and handover management in dense 5G environments. The study emphasizes efficient task offloading during handovers, enabling seamless connectivity by predicting user mobility and optimizing network resources in real-time. While this work provides a detailed taxonomy of data sources (e.g., visual vs. network data), it does not analyze algorithmic complexity or simulation validation tools.

The study by [27] highlights how AI/ML models improve task scheduling in fog computing, particularly in managing large data volumes and mitigating inter-cell interference. These models help in optimizing task offloading strategies to enhance EE and overall network performance. Yet, the paper lacks a detailed classification of AI models per scenario and omits explicit performance benchmarking or simulation environments.

Similarly, [55] discusses distributed learning as a key approach to task scheduling and offloading in heterogeneous networks. The use of distributed ML models not only improves resource management but also ensures data privacy and minimizes latency, which is critical in dynamic and heterogeneous environments. Nonetheless, it does not provide a structured categorization across task offloading, resource allocation, and spectrum domains, nor does it include dataset provenance or toolchain descriptions

2) RESOURCE ALLOCATION AND MANAGEMENT

Resource allocation is a critical area in 5G networks, where AI/ML models are leveraged to manage resources like Bandwidth (BW), power, and computational capacity efficiently.

In [56], ML models optimize EE in 5G networks, focusing on managing power consumption and deploying energy-harvesting solutions in dense networks. The study demonstrates that AI techniques effectively manage dynamic energy resources to reduce operational costs while maintaining performance. However, this work does not provide a systematic classification of AI algorithms, nor does it analyze execution overhead or the use of simulation tools.

Reference [57] investigates how ML enhances Vehicle to Everything (V2X) communications by optimizing resource allocation for data sharing and improving safety in autonomous driving systems. AI/ML dynamically adjusts resource distribution based on vehicle mobility patterns, improving overall resource utilization. Nevertheless, the study primarily focuses on collaborative learning and vehicular environments, lacking comparative performance benchmarks or cross-scenario generalization analysis.

In [58], a comprehensive review of AI/ML applications in Beyond 5G (B5G) and 6G networks is provided. The study examines the role of AI/ML in network automation, resource management, and service performance across various network layers, including RAN and core networks. But this

TABLE 1. Summary of related works' coverage of the topics in this literature review.

Ref, Year	Analyzed Area	AI Learning Type	Dataset Source	Scenario	Simulation Tool	ML Tool	Used Hardware	Runtime
[53], 2020	Network optimization in Proactive 5G Cloud-edge computing framework	✓	✓	✓	×	×	×	✓
[54], 2021	Handover management in 5G/B5G networks	✓	±	✓	×	×	×	×
[27], 2021	5G wireless networks	✓	±	✓	×	×	×	×
[55], 2022	5G and beyond cellular networks	✓	×	×	×	✓	×	±
[56], 2022	Cellular network communications, Computation oriented communications for green communications	✓	×	✓	×	×	×	×
[57], 2022	5G vehicular communications	✓	✓	×	×	×	×	×
[58], 2023	AI/ML in 5G and Beyond Intelligent Networks	✓	✓	✓	✓	✓	×	×
[59], 2024	C-RAN optimization for low-latency networks	✓	×	✓	±	×	×	×
[60], 2024	ML-enhanced Open RAN	✓	±	✓	×	×	×	×
[61], 2024	URLLC in 5G/6G networks	✓	×	✓	×	×	×	×
[62], 2024	Network Slicing for Smart Cities in B5G	✓	×	✓	✓	×	×	×
[63], 2024	Radio Resource Allocation for 5G/B5G	✓	×	✓	×	×	×	×
[64], 2024	Resource Management in 5G/6G Networks	✓	×	✓	×	×	×	×
[65], 2024	AI-empowered network management for 5G/6G	✓	✓	✓	×	×	×	×
[66], 2024	Artificial Intelligence of Things in 5G	✓	×	×	×	×	×	×
[67], 2021	Multiple Access NOMA in 5G wireless communications	✓	×	✓	×	×	×	±
[68], 2021	Software-Defined-Networks and 5G networks	✓	±	✓	×	×	×	×
[69], 2023	AI-driven RAN in 6G networks	✓	✓	✓	×	×	×	×
[70], 2024	Spectrum Sharing Techniques in 5G/B5G Networks	✓	✓	×	✓	✓	×	×
[71], 2020	Channel measurements, modeling and estimation, Physical layer, Network management and optimization in B5G networks	✓	×	×	×	×	×	×
[72], 2020	MNO and Micro-operator in Cognitive radio of 5G/B5G	✓	×	✓	×	×	×	×
[73], 2024	Generative AI for physical-layer channel estimation, equalization, and wireless signal generation	✓	×	✓	✓	×	×	×
This Work	AI/ML for 5G/6G RAN optimization	✓	✓	✓	✓	✓	✓	✓

research falls short of detailing algorithm-specific simulation tools, dataset types, or training methodologies.

Reference [59] provides a comprehensive overview of Cloud Radio Access Network (C-RAN) architectures in B5G networks, highlighting AI/ML's role in optimizing resource allocation and managing ultra-dense network environments. The integration of AI/ML techniques in C-RAN improves both fronthaul and backhaul architectures, enhancing spectral and EE. While technically detailed on C-RAN, it lacks broader classification across AI categories, use-case granularity, and tool validation.

The work in [60] focuses on Open Radio Access Network (O-RAN) architectures, particularly how AI/ML models integrated into the RAN Intelligent Controller (RIC) contribute to optimizing energy consumption. AI/ML models are utilized for resource disaggregation, RAN slicing, and energy optimization in various RAN components, comparable to Central Units (CUs) and Distributed Units (DUs). Although it presents explicit energy models for O-RAN components, it does not cover dataset transparency, training platforms, or multi-task evaluations.

In [61], AI/ML techniques are applied to optimize resource allocation in Ultra-Reliable Low-Latency Communication (URLLC) environments, AI-driven beamforming and interference management techniques are crucial in meeting the stringent reliability and latency requirements of URLLC, especially in applications such as industrial automation and telesurgery. However, it provides limited insights into comparative algorithmic complexity, hardware dependencies, or end-to-end pipeline assessments.

Reference [62] discusses the use of AI/ML for dynamic resource allocation in network slicing, particularly for smart city applications. AI/ML models enable predictive analytics and resource allocation optimization, ensuring that network slicing meets the diverse QoS requirements of smart city services, in particular transportation and emergency response. However, while the survey is rich in application diversity and domain-specific insights, it lacks systematic classification of

AI techniques by scenario and does not evaluate algorithmic complexity or dataset types.

In [63], AI/ML is employed for dynamic radio resource allocation in 5G and B5G networks. The study highlights the integration of AI/ML in RIC to optimize resources such as power, frequency, and BW, thereby improving the flexibility and scalability of network operations. Nevertheless, the work provides limited insight into training procedures, tools, or the reproducibility of the proposed ML frameworks.

The evolution of resource management techniques from 5G to 6G networks is explored in [64], which focuses on network slicing within End-to-End (E2E) frameworks, where AI/ML techniques are critical for optimizing cross-domain resource management and ensuring efficient orchestration across RAN, transport, and core networks. While this survey presents a strong E2E perspective, it lacks granular AI algorithm classification per network function and does not offer simulation or benchmarking analyses.

Reference [65] explores Federated Learning (FL) in resource management, particularly in distributed mobile networks. FL enables network entities to collaboratively optimize resource allocation without sharing sensitive data, thereby improving privacy, efficiency, and facilitating real-time decision-making in distributed environments. Although this work excels in FL-specific mechanisms, it is narrowly focused and does not address broader AI integration or cross-comparison with other learning techniques.

Reference [66] discusses the integration of AI with the IoT, focusing on applications including healthcare, Augmented Reality (AR), and autonomous driving. AI/ML techniques enhance IoT systems by improving decision-making, human-machine interaction, and operational efficiency. While this survey is comprehensive in terms of domain diversity, it treats RAN resource allocation lightly. Furthermore, it omits critical evaluations of the cost, tool usage, and hardware dependency of ML models.

3) SPECTRUM MANAGEMENT

Efficient spectrum management is critical for ensuring the optimal use of available BW in increasingly congested networks. AI/ML techniques have been extensively adopted to optimize spectrum allocation and minimize interference.

In [67], DL models, specifically CNN and RNN, are used to manage spectrum resources in massive-Multiple Input Multiple Output (mMIMO) and Non-Orthogonal Multiple Access (NOMA) environments. AI/ML techniques predict interference patterns and dynamically allocate spectrum resources to optimize system performance and spectral efficiency. Noting this, the survey primarily emphasizes DL architectures across multiple layers but does not categorize spectrum management tasks, benchmark toolsets, or datasets.

Another study [68] emphasizes the use of AI/ML models for congestion control, resource management, and intrusion detection, highlighting their importance in optimizing spectrum usage. The study demonstrates how AI/ML models improve the resilience and efficiency of 5G networks under heavy loads. Yet, it lacks a dedicated breakdown of spectrum-focused tasks and does not analyze specific spectrum-sharing protocols or associated training methods.

Reference [69] focuses on AI's role in transitioning from 5G to 6G networks, particularly in addressing network complexity, resource management, and efficiency challenges. The study highlights applications as intelligent beamforming, dynamic spectrum management, and network slicing. Although comprehensive in the 6G vision, it remains conceptual and omits the classification of AI spectrum algorithms, dataset sources, and execution metrics.

Reference [70] reviews AI/ML techniques for spectrum-sharing in B5G networks, particularly focusing on Cognitive Radio (CR) systems and blockchain-based spectrum allocation. AI-driven models enable dynamic spectrum allocation, improving spectral efficiency and minimizing interference between primary and secondary users. While rich in cognitive radio and regulatory perspectives, this work lacks scenario-specific evaluation of AI models, transparency of datasets, and reporting of simulation environments.

4) CHANNEL ESTIMATION

Channel estimation is crucial for maintaining communication quality in dynamic environments. AI/ML techniques are increasingly used to improve the accuracy of channel estimation in mMIMO, millimeter Wave (mmWave), and NOMA systems.

Reference [72] explores how AI/ML models are employed to address the challenges of channel estimation in high-mobility environments, where frequent handovers and massive connections create highly dynamic conditions. The study shows how AI/ML improves resource orchestration and ensures better channel estimation, improving overall QoS. The work does not offer a task-specific taxonomy for channel estimation methods, nor does it detail the learning tools, dataset types, or training complexities.

Reference [61] discusses the application of AI/ML models in optimizing beamforming and channel estimation for URLLC. AI-based models help ensure high precision in channel estimation, optimizing network performance under stringent latency and reliability requirements. While insightful for URLLC applications, it lacks comparative performance metrics across different AI models and omits training/testing conditions.

In [73], the authors provide a comprehensive survey of the application of Generative AI (GAI) techniques, including GANs, VAEs, normalizing flows, and diffusion models, to physical layer communications, such as channel estimation and equalization. It illustrates how GAI models overcome the limitations of traditional DL models in handling non-Gaussian, non-I.I.D., or rapidly varying channel conditions, enabling tasks like blind equalization and sample-efficient training. While this survey excels in highlighting the potential of GAI in future physical layer designs, it does not explicitly focus on RAN-specific implementations, nor does it benchmark model performance or simulation environments in a structured way.

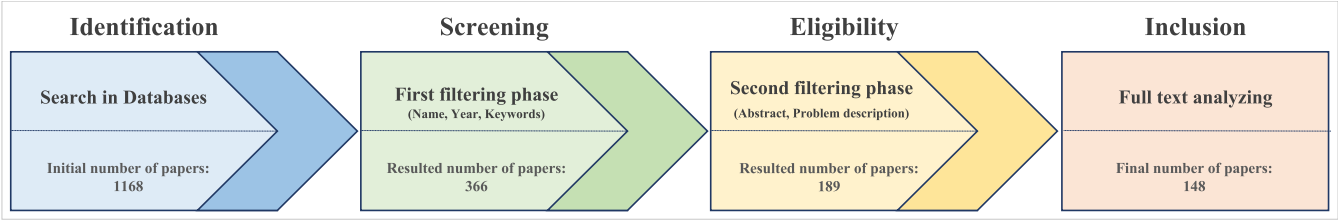


FIGURE 3. Search and screening PRISMA phases.

III. RESEARCH METHODOLOGY

This section presents a methodology based on a systematic literature review following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [74], which ensures transparency and rigor in the identification and inclusion of relevant studies.

A. GAP ANALYSIS OF EXISTING REVIEW EFFORTS

By adopting the PRISMA framework, this survey aims to systematically examine and synthesize existing research on the application and integration of AI mechanisms in 5G networks. The goal is to provide a comprehensive analysis of the benefits and challenges of employing intelligence in communication systems, addressing a significant gap in the literature, as no prior research has fully quantified the costs associated with AI integration.

Although many literature reviews and surveys on AI/ML applications in 5G have been published [53], [56], [67], [68], [72], there remains a gap in providing a comprehensive overview of the datasets, tools, and hardware used. For instance, while [56] reviews efforts towards green communications, it does not specify the hardware or the runtime costs of the experiments. Similarly, [68] surveys AI/ML's role in congestion control and intrusion detection, identifying future challenges like the trade-off between computation load, accuracy, and interpretability, but lacks insights into real data usage versus simulations.

Several works, including [59], [60], [61], offer valuable insights into AI/ML applications in 5G networks but lack practical details, in particular the datasets, hardware specifications, and runtime costs of their models. For instance, [60] addresses O-RAN energy optimization but does not cover the real-time trade-offs between accuracy and efficiency. Similarly, [61] discusses URLLC resource allocation but omits key implementation details, such as simulation environments and hardware constraints. Future research should focus on these missing aspects to ensure more practical and scalable AI/ML deployments in 5G networks.

When stating future challenges, they identify several problems that need to be tackled. Some of these challenges include the trade-off between computation load and accuracy, the limitations of simulations that only provide approximations, the scarcity of real data, and the need for testbeds. Even so, to the best of the authors' knowledge, no survey or literature review has quantified the percentage of papers that use real data, simulations, or testbeds. Furthermore, the literature lacks an analysis of the simulation and ML tools used, the hardware employed for testing the models, and the runtime cost of these models.

The subsequent sections outline the detailed search strategy, inclusion and exclusion criteria, and the data collection process, each of which is structured according to the PRISMA methodology.

B. PRISMA STRATEGY

PRISMA involves four main stages: identification, screening, eligibility, and inclusion [74]. The PRISMA flow chart in Figure 3 illustrates the corresponding phases followed across the selected databases.

1) IDENTIFICATION

The Identification step corresponds to the search strategy adopted to identify relevant literature. Key articles were retrieved from leading databases, including IEEE Xplore, Scopus, Springer Link, ACM Digital Library, Wiley, and MDPI. These papers presented current approaches, experimental applications, and results from ML techniques within the RAN domain for 5G networks. The search was conducted using several keywords and related alternatives, focusing on terms similar to '5G', 'RAN', 'machine learning', 'artificial intelligence', and 'deep learning'. Some example queries used are:

- ('5G' and 'RAN' and 'Beyond 5G') or ('AI' or 'ML')
- ('AI' or 'ML') and ('5G' or '5G RAN' or 'power consumption' or 'latency optimization')
- ('Deep learning' and 'energy efficiency') or ('task offloading' or 'resource allocation')
- ('ML' and 'B5G' and 'estimation') or ('bandwidth control' or 'traffic prediction')
- ('DL' and '5G') or ('vehicular communication' or 'minimize latency')
- ('ML' and '5G') or ('Multiple Input Multiple Output (MIMO)' or 'mmWave' or 'channel estimation')
- ('DL' and '5G') or ('MIMO' or 'Orthogonal Frequency Division Multiplexing (OFDM)' or 'spectrum optimization')

The raw queries, executed without the inclusion/exclusion filters, yielded more than 10,000 articles spanning a wide range of disciplines and unrelated domains. These were subsequently narrowed through focused keyword logic and refined constraints, resulting in an initial relevant set of 1,186 papers.

2) SCREENING

The screening phase involved the initial filtering of articles, where the selection of articles followed specific criteria to ensure objectivity and relevance, as listed below:

- The research was published starting from 2019.
- The research was published in English.
- The research focuses on RAN in 5G and future networks.
- The articles address, propose, or innovate AI/ML techniques, methodologies, and algorithms to solve critical challenges in 5G and beyond networks.
- Detailed descriptions of the employed parameters, variables, initial values, and the structure of the proposed AI/ML algorithms are provided.
- The publications were published in high-impact SCI-indexed journals, or presented at reputable conferences with core B or above [75], included in major databases (i.e., IEEE Xplore, Scopus, MDPI).

The exclusion criteria were as follows:

- Research outside the context of AI/ML solutions for RAN in a 5G environment.
- Research not directly aligned with the objectives of this study.
- Early versions of research based on the same data, exploring the same topics.
- Research not written in English.

Applying these criteria, the initial pool of papers was reduced from 1186 to 366.

3) ELIGIBILITY

This stage involved a more detailed assessment of the filtered articles, where the selection process required an analysis of the abstract to validate if the works met the exclusion and inclusion criteria outlined in III-B2. In some cases, the review extended to the introduction for further clarity.

4) INCLUSION

In the final phase, full texts of the remaining 189 articles were examined to identify those that provide substantial contributions to the field. A total of 148 articles were retained and analyzed systematically. Each was evaluated across several dimensions, including:

1) Network framework: refers to the area of the RAN network, in which the subject of the work was studied.
2) AI algorithm type: It is important to describe the AI algorithm integrated into the proposed solutions in each section of this study. As described in section II-B, there are several algorithms, and each has pros and cons. Besides, the ratio of DL is compared to shallow learning algorithms.
3) AI tools/libraries: As is known, the general AI/ML method evaluation includes three steps, namely training, testing, and validating. The used tools/libraries to implement the proposed AI/ML methods were noted as a part of the essential components to obtain the full solution.
4) Scenario realization: indicates the steps followed to generate the described experiment scenario. The realization of the experiment can occur through three different scenarios:
   - Simulation: Network simulators are utilized because communication networks have become too complex for traditional analytical techniques to provide precise knowledge of system behavior. A network simulator is a software program that mimics real-world network activity and predicts the performance of a computer network or a wireless communication network.
   - Testbed: that combines a set of composite abstractions of systems that are employed to mimic systems' components and interactions to understand the fundamentals of the real system better. They are made of prototypes and subsets of real system components that are gathered to shed light on the operation of a system element or elements. The testbed's main advantage is the ability to virtually expand real production settings with equipment that is either not yet available or not yet acquired.
   - Real environment: to validate a proposed solution in a real-world environment, the experiment should be controlled and systematic. This is achieved by following several steps, including defining the problem and hypothesis, performing tests, analyzing the results, and concluding with findings and recommendations to ensure reliable results.
5) Dataset: One of the main points is describing the method used to generate the dataset. In this study, we divided the methods of generating the dataset into three main types, as described below:
   - Synthetic: The dataset is generated according to a mathematical distribution or a customized distribution.
   - Simulated-based: herein, generating the dataset relies on well-known simulators, i.e., Network Simulator (NS-3) and MATLAB.
   - Real data: This type of dataset is extracted from a testbed, production system, or network vendor and contains operational data.

IV. SYSTEMATIC ANALYSIS

This section provides a comprehensive comparative analysis of existing research across the four functional domains introduced earlier in Section II-A, namely task offloading and scheduling, resource allocation and management, spectrum management, and channel estimation. To ensure consistency and depth, each work is evaluated using a uniform set of criteria derived from a systematic methodology. This approach enables a structured interpretation of how AI techniques contribute to performance improvement across various RAN functionalities.

A. TASK OFFLOADING/SCHEDULING APPROACHES ML-BASED

In 5G networks, task offloading requires optimizing offloading decisions and resource utilization, allowing end-users to

achieve better service by sending the computation-intensive tasks to outsourced entities, including fog nodes, mobile Edge computing nodes, and cloud nodes. These nodes execute tasks to lighten the workload of the end-user [76].

In recent years, numerous research papers on offload computing have emerged, considering the most recent AI techniques to predict where to execute tasks based on environmental variables. In this sense, adaptive schemes are highlighted, mainly using RL techniques. In the 5G and B5G networks, resource optimization [77], minimizing the average latency, and EE [78] will be discussed in detail as a part of the optimization objectives achieved based on joining AI methods with task scheduling techniques. This integration improves dynamic resource allocation in these networks [79].

1) ENERGY EFFICIENCY

Improving the efficiency of task offloading in terms of EE is a critical problem to be solved, and the key objective is to propose a task offloading scheme able to minimize the overall energy consumption, besides considering essential parameters suchlike sub-carriers, power, and BW for offloading to guarantee minimum energy consumption along with satisfying capacity and delay requirements [80].

The authors in various studies endeavor to minimize the energy consumption of task offloading through the application of DRL algorithms. In the context of a Software-Defined industrial fog network, the authors in [81] proposed a DRL-based service provisioning strategy that leads to a reduction in system energy consumption. The proposed DRL-based strategies significantly reduce power consumption compared to reference methods, although they may involve longer convergence times, depending on the hyperparameter settings. In [82], the authors proposed a DRL method for efficient job scheduling and resource allocation. DRL allocates compute-intensive work to servers and minimizes energy consumption while maintaining job processing performance. In [83], a new DRL control scheme was proposed, utilizing partial download based on resource allocation across multiple 5G Radio Access Technology (RAT) and Multi-access Edge Computing (MEC) systems. The User Equipment (UE) in the proposed method makes ideal decisions for offloading, and the MEC server utilizes DRL technology to dynamically adjust server resources in response to offloading requests from various UEs. In [84], the authors employ a combination of DRL and RL to enhance resource utilization and performance isolation between segments, specifically utilizing DRL for large-time-scale resource allocation. In contrast, RL is used to perform online resource scheduling for tackling small-time-scale network dynamics, including inaccurate predictions and unexpected network states.

The study [85] proposes a DRL-based task scheduling framework for cloud computing systems, focusing on optimizing EE and task handling. The approach improves EE and Service-Level Agreement (SLA) adherence, demonstrating practical effectiveness for task scheduling in cloud environments. The authors of [86] proposed a scheduler that enhances EE in O-RAN by optimizing multipath data transmission through dynamic resource allocation, thereby reducing power consumption. Utilizing a hybrid DRL model and a Transformer-based Throughput prediction model, the scheduler enhances bandwidth aggregation, reduces bottlenecks, and promotes green communication by intelligently managing data flows.

In contrast, SL is employed in [87] to predict energy consumption in a fog computing environment. In this case, the concept of task offloading is used based on the size and number of tasks, as well as the Central Processing Unit (CPU) cycles required to execute each task.

Other works explore Optimization techniques and hybrid models, such as neuro-fuzzy systems and rule-based strategies, to enhance EE in distributed computing environments.

In [88], a secure computing offloading scheme for the Fog-Cloud-IoT environment is presented. It utilizes a Neuro-Fuzzy model [89] to protect data at the smart gateway and employs an optimization process to select the optimal fog node for offloading. If the fog node cannot handle the workload, it sends it to the cloud as either sensitive or non-sensitive data. The optimal fog node selection is based on maximizing two metrics, Available Processing Capacity (APC) and Remaining Node Energy (RNE), using the RL strategy. In [90], the authors optimize the energy and time of the MEC system using a multi-level offloading technique that balances workload energy in the IoT and MEC in the 5G network. The tasks are divided and offloaded to the MEC server and core cloud for execution, and only a small number of tasks are assigned to mobile devices.

This study [91] introduces an energy-efficient task offloading and load-balancing framework for IoT systems in fog-cloud networks. The system employs a Neuro-Fuzzy scheduling mechanism combined with a binary linear-weight algorithm to optimize energy consumption and latency.

2) MINIMIZE LATENCY

The 5G wireless networks paved the way for the advent of many new applications and services, most of which are latency-sensitive, in particular Virtual Reality (VR), AR, and vehicular communications. Tasks offloading is one of the basic optimization methods to minimize the latency of all end-users in the system, considering the tasks' time demands [92].

Several studies apply SL techniques to minimize latency in task offloading by enabling predictive decision-making based on system state and task characteristics. The authors in [93] propose an ML-based offloading mechanism that can automatically offload traffic from congested cells to less congested ones in Heterogeneous Ultra-Dense Networks (HUDNs). This scheme enables the use of shallow SL for feature learning to determine the link quality, thereby assisting the user in accessing a stable cell. This allows the network to maintain a relatively stable state of load balancing for an extended period. In addition, RL can be used to decide the associated cell for each user, considering not only the

communication capacity but also the load of the candidate cells for the user association decision.

In [94], an AI-based framework for fast task execution decision-making in delay-sensitive IoT applications within edge computing networks is proposed. The system utilizes SL to expedite task offloading decisions by optimizing factors such as task execution deadlines, energy consumption, and server capacity. The framework improves the speed and efficiency of task offloading in heterogeneous IoT environments.

A growing body of works leverages DRL to dynamically reduce latency in complex and evolving edge and fog computing environments. One such work [95] employs DRL with RNN to design an online predictive offloading framework that aims to reduce long-term expenditures related to latency and energy usage. Tasks are dynamically offloaded to the ideal edge server by anticipating task characteristics and edge server workloads, besides avoiding generating network congestion and server overload. The work [96] proposed a model to solve computing offload for Edge Computing Server (ECS) heterogeneous collaborative computing. Considering multi-tasking, the heterogeneity of edge subnetworks, and the mobility of edge devices, the proposed algorithm can learn the network environment and generate computational offload decisions to minimize task delay and maximize task completion. In a later study [97], an IoT environment was connected to a distributed edge computing system. This system provides communication and computation offloading for IoT devices in scenarios involving both static and dynamic system models. To minimize task queuing delay and transmission costs between IoT devices and edge nodes, a DRL algorithm combined with FL was applied, allowing tasks to be performed locally on the IoT device or offloaded to the edge node. This decision is based on Channel State Information (CSI), task queue state, and energy queue state. Executing the task on the edge node requires considering the radio channel quality between the IoT device and the edge node, as it directly affects the transmission rate.

The authors in [98] present a DRL-based distributed task offloading framework for optimizing task offloading decisions in Edge-Cloud Computing (ECC). The approach is scalable and adaptable to dynamic workloads, demonstrating significant improvements in task execution efficiency in Edge-Cloud environments. The framework offers a more stable and efficient offloading mechanism compared to conventional methods.

Other proposals adopt optimization-based methods, including hybrid techniques, to reduce latency by making precise scheduling and offloading decisions. The work [99] aimed to minimize latency while achieving EE in the Industrial Internet of Things (IIoT) environment, assisted by mobile edge computing. The authors used a task scheduling strategy based on an optimization method. An Effective Dynamic Load Balancing (EDLB) technique was introduced in [100] that uses a modified optimization method and a CNN-based classifier. The classifier categorizes fog servers as suitable or unsuitable, while the optimization algorithm assigns incoming processes to the most appropriate server to minimize task execution time.

In [101], the authors propose an optimization-based algorithm for latency-aware prioritization and task offloading strategies in fog-cloud IoT environments, utilizing fuzzy logic. Tasks with stringent latency requirements are prioritized and mapped to computationally capable nodes to reduce transmission and service delays, which leads to a reduction in average latency.

3) RESOURCE OPTIMIZATION

The optimization of resources in a 5G wireless environment, where URLLC is a crucial factor in massive Machine Type Communication (mMTC), has become increasingly important due to the scarcity of radio resources. Consequently, efficient resource optimization methods have become a mandatory tool to meet demands for low latency, reliability, density, and BW [102].

Several studies have applied RL strategies to improve resource scheduling and utilization in diverse networking environments. In [103], the authors proposed an RL-based scheduler for a multi-resource cluster, focusing on complex cluster operating conditions and diverse resource requirements. In [104], RL was utilized to achieve optimal resource utilization, considering the reduction of total computing time for tasks in an IoT environment through FL. The proposed method, when applied at the edge node, determines whether the tasks should be offloaded to a server based on the availability of resources. In [105], RL was utilized to optimize the placement of resource allocation functions and decisions based on service requirements.

In the paper [106], the authors propose an optimization hybrid algorithm that combines an RL algorithm with energy-efficient task scheduling for cloud-based networks. The method aims to minimize makespan, cost, and resource utilization while maximizing Virtual Machine (VM) throughput.

Other studies presented solutions that utilized DRL as [107], [108], and [109].

In [107], a method was proposed to manage distributed scheduling in a small set of cells by dynamically selecting schedulers through the use of DRL and a Genetic Algorithm (GA). Based on the current state of the network and considering the available resources, this dynamic scheduler selection system was able to identify scheduler distributions that improved the proportion of users meeting QoS requirements. Study [108] introduces a divisible task offloading strategy for multiuser, MEC systems. By utilizing a DRL with prioritized experience replay, the framework optimizes task scheduling and resource allocation among multiple servers and users. The approach is designed to enhance the system's efficiency in managing divisible tasks, leading to significant improvements in resource utilization and reduced latency. As for [109], the study proposes a priority-based resource scheduling algorithm using DRL for edge computing networks, focusing on efficiently allocating resources for

high-priority tasks. The framework uses heuristic algorithms to manage resource allocation and optimize task scheduling performance.

Finally, some works rely on optimization-based approaches to enhance resource efficiency through multiobjective scheduling and load balancing techniques. In [110], the authors introduce a multiobjective optimization-based task scheduling algorithm for resource optimization in cloud-fog computing. Designed to improve resource allocation and minimize both transmission delay and energy consumption. The algorithm balances the computational load across fog and cloud layers, dynamically allocating tasks based on network conditions and resource availability. The model employs Harris Hawks optimization to provide a set of Pareto-optimal tradeoff solutions.

4) COMMENTS AND DISCUSSION

The analysis of AI techniques in the Task Offloading/ Scheduling subsection reveals the division into three main categories: energy efficiency 36.1%, latency minimization 36.1%, and resource optimization 27.8%. As observed from the distribution in Figure 4, the group as a whole exhibits a preference for DRL, which stands out prominently with the highest share across all three categories.

At the subclass level, starting with EE, DRL leads significantly, accounting for 38.5% of the methods employed, underscoring its ability to adapt dynamically to optimize energy consumption. RL, optimization methods, and Neuro-Fuzzy systems each account for 15.4%, reflecting their robust contributions to energy-saving strategies. Deep Q-Learning and SL share an equal portion (7.7% each), demonstrating their potential in targeted energy-efficiency tasks.

In the minimizing latency category, DRL again dominates with 30.8%, followed by optimization methods at 23.1%. Fuzzy Logic, SL, CNN, RL, FL, and RNN all converge closely at approximately 7.5% each. This clustering indicates broader experimentation with a variety of methods to address latency, reflecting the complexity of latency-sensitive applications like AR/VR and vehicular communication.

Finally, in resource optimization, DRL is the predominant method, accounting for 50% of the approaches. The other techniques, including optimization methods and RL, each claim a nearly equal share (20%). FL with a 10% ratio is less prominent but still contributes to addressing specific resource allocation challenges.

Regarding tools and libraries, TensorFlow and PyTorch dominate with 23.5% and 17.6%, respectively. Other tools like Scikit-learn also find usage at 5.9%. Notably, 52.9% of the surveyed papers did not explicitly mention the tools used, indicating a lack of transparency or standardization in reporting AI toolsets. For simulation platforms, Python and

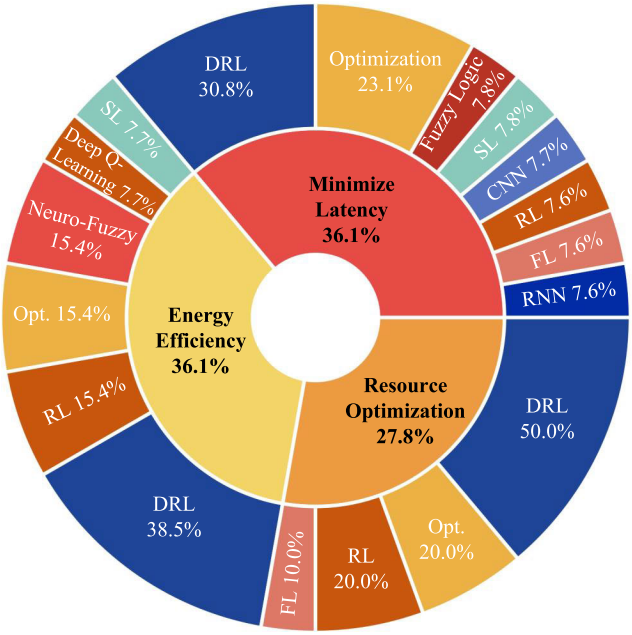


FIGURE 4. Distribution of AI algorithms for task offloading and scheduling.

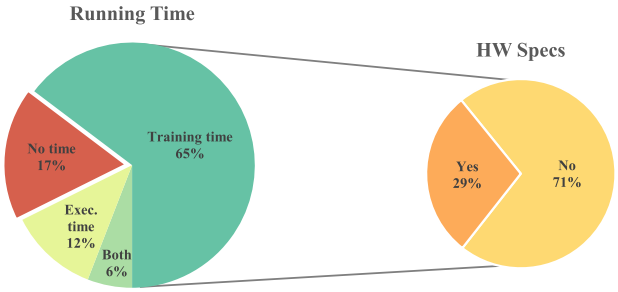


FIGURE 5. Execution time and hardware specifications for task offloading solutions.

Java each capture 11.1% of use, while NS-3 and NetworkX occupy 5.6%. However, a substantial proportion (55.6%) of studies omitted details about their experimental environment, which limits reproducibility.

In terms of the time consumed and the hardware specifications used, the majority of articles provided data on either training time (65%) or execution time (12%), while only 6% disclosed both metrics. This reflects a growing awareness of the importance of computational performance in AI-based research. However, as illustrated in Figure 5, of the 83% that mentioned the needed time, only 29% of the studies detailed hardware specifications, highlighting an area for improvement in reporting standards.

The lack of consistent documentation on hardware directly impacts the comparability and scalability of the proposed AI solutions.

Table 2 provides more specific information about the surveyed articles, including more details about the experiment scenario, the tools utilized to carry out the experiment or train

[1] https://doi.org/10.1109/JIOT.2021.3073034
[2] https://www.sciencedirect.com/science/article/pii/S0743731500917143
[3] https://doi.org/10.1109/ICC.2015.7248881
[4] http://dx.doi.org/10.1007/s10586-018-2414-6
[5] http://dx.doi.org/10.3390/data3040038

TABLE 2. Summary of AI-based task offloading and scheduling solutions.

Ref, Year	Network Domain	Scenario Realization	Scenario validation tool	System Model /Topology	AI Algorithm Type	ML Tool	Dataset Source	Used HW	Run-time
				Energy Efficiency					
[88], 2019	Fog Cloud	Simulation	NS-3, Java	1 smart GW, 5-10 IoT MT 5 FN,1 H-cloud server	Neuro-Fuzzy, Optimization, RL	X	Synthetic	✔	✔
[84], 2019	5G RAN	Simulation	Python	1 -20 network slices/radio blocks, prediction window 3600s	DRL, RL	TensorFlow	Synthetic	✘	✔
[82], 2019	Datacenter	Simulation	X	1 smart GW, 5-10 IoT MT 5 FN,1 H-cloud server	DRL	Custom code	Testbed	✔	✔
[87], 2021	Fog	Simulation	X	Requests size 0.5 to 16 MB FN serves 50 to 100 requests	SL	X	Synthetic	✘	✔
[90], 2022	Mobile Edge Computing	Testbed	Testbed	500 tasks offloaded into MT, SBS, and main BS (MBS)	Optimization	Scikit -learn	Synthetic	✘	✔
[83], 2022	MEC	Simulation	X	1 BS, 8 SBS, 30 ENs Max 40 UE per BS	DRL	PyTorch	Synthetic	✘	✔
[81], 2022	Software-Defined Industrial Fog net.	Simulation	X	1 SC, 5-10 UE, 4 FN Task size 2-20 MB	DRL	TensorFlow	Synthetic	✔	✔
[91], 2024	IoT, Fog-Cloud	Simulation	iFogsim	Multi-user environment, distributed edge servers	Neuro-Fuzzy	X	Simulated	✘	✘
[85], 2024	Cloud	Simulation	Cloudsim	35 VMs, 1000 tasks, across 10 datacenters	Deep Q-Learning	X	Simulated	✔	✘
[86], 2024	O-RAN	Simulation	X	O-RAN with Multipath Scheduler	DRL	X	Testbed	✘	✘
				Minimize Latency					
[93], 2019	HUDNs	Simulation	X	1 BS (a MeNB, 3 SeNBs), 20 UEs around the BS	RL	X	Simulated	✘	✘
[97], 2019	IoT, Edge	Simulation	X	Sensing data collected by 15 IoT devices sent to 6 edge nodes for DRL training, then comparing DRL and FL	DRL, FL	X	Synthetic	✘	✔
[96], 2020	Edge Cloud	Simulation	NetworkX	3 edge devices, 2-8 wireless / 1-4 wired edge servers	DRL	TensorFlow	Synthetic	✘	✔
[99], 2021	Edge	Simulation	Custom code/ Java	50-250 devices, 10 MEC server	Optimization	X	Synthetic	✘	✔
[95], 2022	IoT Fog	Simulation	X	500 tasks, 5 edge servers 50 terminal devices	DRL, RNN	X	Real data	✘	✔
[100], 2022	IoT Fog	Simulation	X	150 tasks, 200 Fog servers	CNN, Optimization	X	Simulated	✘	✘
[94], 2023	IoT, Edge	Simulation	X	10 smart IoT devices 2 UAV, 6 edge servers	SL	X	Simulated	✔	✘
[101], 2023	IoT, Fog-Cloud	Simulation	iFogSim	IoT-fog-cloud, 3-layer model	Fuzzy Logic, Optimization	X	Real data	✘	✘
[98], 2024	IoT, Edge-Cloud	Simulation	Custom code/ Python	2 - 3 mobile devices 2 edge servers and cloud resources	DRL	Keras	Simulated	✘	✘
				Resource Optimization					
[103], 2019	HetNet Cluster	Testbed	Testbed	HetNet cluster with 3 levels, a cluster has 3-10 varied machines	RL	PyTorch	Simulated	✔	✘
[107], 2020	5G RAN	Simulation	X	7 cells in a hex grid layout, different scenarios over small periods, increase from 1 -100 TTI with static users	Optimization, DRL	X	Real data	✘	✔
[105], 2021	5G RAN	Simulation	NS-3	4 BSs, 40 UEs randomly distributed, 12 subcarriers, each has 15 KHz subcarrier spacing and 14 symbols per subframe	DRL	PyTorch	Synthetic	✘	✘
[104], 2021	IoT Edge	Simulation	X	3 scenarios with 6, 10, 14 tasks, between 2 and 4 edge nodes	FL, DRL	PyTorch	Synthetic	✘	✔
[108], 2023	MEC, Cloud	Simulation	X	Divisible tasks 100 tasks, 30 used device 5 edge servers	DRL	X	Simulated	✘	✘
[106], 2023	Fog, Cloud	Simulation	CloudSim	200 - 1000 tasks, 100 VM	RL	X	Simulated, Synthetic, Real data	✔	✘
[109], 2023	IoT, Fog-Cloud	Simulation	CloudSim	10 - 100 tasks, many user devices Fog nodes, a Cloud	DRL	X	Simulated	✘	✘
[110], 2024	Fog-Cloud	Simulation	MATLAB	One cloud, 3 to 40 fog nodes, 50 to 410 tasks	Optimization	X	Simulated	✔	✘

the methods, and the types of the tested datasets, with 11.7% of the datasets extracted from actual working organizations and the majority, 64.7% of datasets were synthetic, generated as a result of costumed experiments. Only 17% of the datasets were generated using simulation tools.

B. RESOURCE ALLOCATION AND MANAGEMENT STRATEGIES ML-BASED

In networks, resource management poses a significant challenge, particularly when the number of connected users or intelligent devices is vast. Efficiently managing resources,

e.g., time slots, frequency bands, orthogonal codes, etc., taking into account variable traffic loads and wireless channel fluctuations is a crucial task [111]. AI techniques enable the utilization of previous and current knowledge about the system, the network, the users, and the parameters to make predictions and inform control decisions in this sense.

1) ENERGY EFFICIENCY

One of the primary objectives of resource allocation is to achieve EE through various techniques, including network planning and development, energy collection, and radio resource allocation [112]. These papers collectively explore advanced energy-efficient resource allocation strategies for 5G and B5G, leveraging machine learning techniques to optimize network performance.

Several works utilize optimization techniques to enhance EE by optimizing resource allocation and task scheduling. In [113], an algorithm based on optimization is proposed to improve the EE of Fog-RAN. Specifically, user association, resource allocation, and caching strategy are formulated as a non-convex optimization problem and then transformed into a convex problem, which is solved by a proposed optimization algorithm.

In [114], the main objective is to ensure secure communication in the IoT network and improve EE in the IoT network. To do so, a novel power allocation intelligent optimization algorithm is proposed, which utilizes swarm intelligence and adaptive weights to update the position information.

In [115], SL was utilized for resource management in a specific part of a RAN, particularly for Base Station (BS) activation and deactivation, aiming to reduce energy consumption while providing sufficient capacity. Conservative strategies reduce BS hardware failures, while newer dynamic devices enable frequent BS switching for improved EE.

A growing number of contributions utilize RL to optimize EE by dynamically adjusting resource use in response to network conditions. The authors in [116] present an adaptive task offloading algorithm for Vehicular Fog Network (VeFN)s that addresses and exploits mobility through opportunistic computation offloading and task replication.

In [117], RL is used to improve EE through intelligent antenna switching, i.e., disabling some of the antennas installed at a 5G mMIMO BS when there are few UEs within the cell area. Expanding further into the concept of EE, [118] examines the challenges associated with ultra-dense 5G network deployments. Using RL, the study optimizes both the placement of access points and resource allocation to maximize EE. This work is crucial for mitigating the carbon footprint of dense 5G infrastructure, highlighting the importance of environmentally sustainable network designs. Shifting toward the future of wireless networks, the paper [119] explores EE in 6G systems. The authors propose a hybrid RL model that optimizes both energy and spectrum resources in distributed 6G systems. This methodology, which integrates both central and distributed resource allocation schemes, aims to minimize energy consumption while ensuring performance benchmarks are met.

Some approaches combine RNN with other learning techniques for predictive energy-efficient resource management. In a distributed 5G RAN environment with Self Organized Network (SON) functionalities, [120] proposes an RNN-based solution to achieve EE based on edge intelligence. The applied scenario had two RAN use cases, the first one is a handover decision, and the second is the selection of the initial Modulation and Modulation and Coding Scheme (MCS). The authors advocated a gossip learning paradigm that enables training by peer-to-peer information exchange, in addition to evaluating the proposed solution in realistic scenarios.

The proposal in [121] builds a dynamic traffic model using an RNN and employs DRL with convex optimization to allocate user power and bandwidth in a 5G C-RAN. Based on the multidimensional data, the RNN is used to build a dynamic traffic model of the multicast service in space-time, providing a basis for further allocation of network resources. Based on the prediction results and the condition of satisfying the changing requirements of users, an optimization model is built to minimize the power usage of the Remote Radio Head (RRH) while taking into account the QoS constraints of users. DRL is then combined with a convex optimization method to complete the allocation of users' power and BW resources.

Other studies apply CNNs for predictive energy-saving strategies. In the 5G network, the authors in [122] presented a CNN model to improve the prediction performance and achieve faster prediction times. The proposed method considers user mobility as spatial information for forecasting temporal mobile traffic. With the ability to accurately predict when users move and turn the Access Point (AP) into different modes (normal mode or low-power mode) ahead, the network can save energy and achieve satisfactory QoS.

An extensive body of research leverages DRL for energy-efficient resource allocation, task offloading, and power optimization. The authors of [123] present a DRL-based computational offloading algorithm for multilevel vehicular edge-cloud computing networks. The objective is to minimize energy consumption and task execution time. The simulation results show the computational cost of the proposal. Although it compares the performance of the proposed model for multi-tier vehicular cloud computing networks with three different policies, it does not compare with other offload algorithms. The authors of [124] utilized DRL to optimize binary and partial downloading, aiming to jointly determine optimal decisions for downloading and resource allocation that maximize the number of bits computed while minimizing power consumption in MEC environments. The results of the experimentation show an improvement in terms of computed bits, total transmission power, and efficiency. In [125], a DRL-based dynamic resource management algorithm is presented in order to minimize the long-term average delay of the tasks for MEC in IoT.

In [126], a DRL algorithm is proposed to maximize the overall EE of communication and computing within MEC servers and sensors by co-optimizing the allocation of channels and computing resources within an IoT system related to emotional computing.

Paper [127] introduces an energy-efficient resource allocation method specifically tailored for 5G networks, utilizing DRL. This method is designed for edge computing and industrial IoT scenarios, aiming to minimize energy consumption while meeting the stringent QoS requirements. The proposed DRL-based approach maximizes EE by optimizing the transmission power of base stations.

Other neural approaches, such as Deep Belief Networks (DBNs) and DNNs, have also been proposed to enhance EE in communication systems. Using a DBN [128], an approach to address the resource allocation problem for a Multi-Carrier Non-Orthogonal Multiple Access (MC-NOMA) system downlink enabled for Simultaneous Wireless Information and Power Transfer (SWIPT) with the scheme Pattern Division Multiple Access (PDMA) was presented at [129]. The objective is to minimize the total transmitting power of the system while satisfying the quality-of-service requirements of the users based on joint downlink resource allocation.

In [130], a DNN was utilized to develop a framework that integrates with state-of-the-art computing devices for automated monitoring and diagnosis, making it particularly useful for real-life healthcare systems, such as those addressing heart disease. The objective of this proposal is to minimize energy consumption, execution time, and comply with network BW and latency, to maximize the performance of healthcare systems in IoT/Fog.

Together, these studies present a comprehensive view of how RL and other ML techniques can be applied to optimize resource allocation, reduce energy consumption, and ensure sustainability in 5G and future 6G networks.

2) MINIMIZE LATENCY

The development of 5G networks has enabled many latency-sensitive applications; consequently, reducing system latency is crucial. The works addressed below explore latency minimization in 5G and 6G networks, using AI to optimize resource allocation and management for low-latency communication.

Several works leverage SL to reduce system latency by enabling accurate, low-latency predictions in network environments. In [131], a novel intelligent multimedia data segregation scheme using SL in the fog computing environment is presented. The objective of the article is to minimize latency in e-healthcare through fog computing, thereby sending data to the end-user in the shortest possible time.

In [132], a new network traffic classifier agent was proposed for the multi-agent-based network automation of the network management system (MANA-NMS) architecture [133], using a model of SL as a cognitive component of the agent.

Other works apply RL to optimize decision-making for latency-sensitive tasks in dynamic and distributed environments. In [134], a framework using RL is proposed for the dynamic display of radio resources in heterogeneous virtualized Radio Access Networks (vRANs), which selects the best available link and transmission parameters for packet transfer, in order to meet the requirements of the Key Performance Indicator (KPI)s.

In a smart city environment, cooperating with Vehicular Fog Computing (VFC), [135] proposed an RL algorithm combined with heuristic information, aiming to minimize service latency for vehicular applications. The proposed solution tracked the movement and parking patterns of vehicles in a smart city, utilizing parked vehicles as edge servers to allocate available computing resources and optimize service latency.

Further extending these ideas, [136] addresses latency minimization in cloud-aided small cell networks. The authors propose a joint caching, scheduling, and resource allocation technique that leverages RL and spectral clustering to optimize content placement and minimize queuing latency. The work highlights the relevance of RL-based approaches and introduces a solution for reducing convergence time in latency-aware resource management for 5G applications.

A substantial body of work employs DRL to dynamically manage computation offloading, caching, and scheduling, thereby minimizing latency across various 5G/6G architectures.

In [137], a resource allocation scheme based on DRL was proposed to improve content distribution in a Fog Heterogeneous Radio Access Network (F-RAN). The DRL policy is designed to make cooperative cross-layer routing and caching decisions for incoming content requests based on request history information and available network resources on the system.

In [138], the metric called Perception-Reaction Time (PRT) is presented, which reflects the time consumption of safety-related applications and is closely related to efficiency and road safety. Additionally, DRL is employed to design an optimal online resource allocation scheme. A time-constrained resource allocation problem was simulated in vehicular fog road architecture. Paper [139] proposes a DRL-based joint computation offloading and resource allocation scheme to minimize latency and increase throughput for low-latency fog radio access networks. DRL controller intelligently decides whether to process the generated computation task locally at the device level or offload the task to a fog access point or cloud server and allocates an optimal amount of computation and power resources based on the serving tier. The authors in [140] implemented an Software Defined Networking (SDN) controller with a DRL algorithm that can route the requests and adapt to different mobile edge computing environments. The proposed algorithm uses a resource allocation scheme that can adaptively allocate both computational and network resources to minimize the average service time and achieve resource allocation load balancing for the edge servers.

Paper [141] applied DRL and Multitask Learning framework to allocate the resource of mobile edge computing, where the proposed framework considers network states and task characteristics, intending to maximize the performance over latency and power consumption.

The authors in [142] proposed a resource allocation strategy based on DRL in an edge cloud computing environment to address the high volume of data transmitted in 5G networks, which causes channel congestion and impacts the real-time performance of communication. The presented download strategy focuses on reducing the average response time of system tasks and total power consumption.

The paper [143] introduces a DRL-based approach specifically designed to minimize latency in URLLC and eMBB services. The method utilizes a hierarchical RL framework to dynamically optimize scheduling and resource allocation, leading to a substantial reduction in transmission delays and overall network latency compared to traditional approaches.

Building on this, the paper [144] addresses latency reduction in 6G networks using the O-RAN architecture. It introduces the RIC for resource management and proposes a DRL method for allocating resources between near-real-time and non-real-time RICs. This complements the hierarchical reinforcement learning framework in [143], broadening the set of strategies for latency minimization in next-generation networks.

The authors in [145] propose a transformer-assisted mobility prediction and decentralized deep deterministic policy gradient-based DRL framework for resource allocation in MEC-enabled smart city networks, focused on minimizing latency through effective load balancing, task migration, and resource allocation. The system dynamically allocates tasks and resources based on predicted mobility patterns and current resource availability, enhancing system efficiency within smart city environments.

Other approaches integrate DNNs to achieve low latency by distributing workloads and predicting optimal resource configurations. The authors in [146], propose data flow and distributed DNN-based low latency IoT-Edge computation model is presented for the big data environment. The result is a reduced network, service, and overall latency by distributing the process or workload at the Edge level.

Paper [147] handled the critical time constraints in the IoT environment where the proposed solution is built on a scalable resource allocation framework assisted by DNN mechanism to achieve the balance between users' density and requests regarding the available resources, leading, in turn, to oblige with time constraints of the tasks. In [148], a computation-offloading optimization problem was solved using a DNN to minimize the total transmit power of IoT devices while meeting the system's task latency requirements.

Another study utilizes FL to achieve low-latency communication in distributed vehicular networks. The authors in [149] propose a novel distributed approach based on FL to enable ultra-reliable and low-latency vehicular communication networks by reducing the average power consumption through a power and resource allocation framework.

Together, these papers present a comprehensive set of strategies for optimizing resource allocation and minimizing latency across various communication scenarios, ranging from URLLC in 5G to latency-sensitive applications in 6G networks, all driven by advanced AI techniques.

3) RESOURCE OPTIMIZATION

In a dynamic network environment, intelligent traffic prediction and monitoring methods play a crucial role in enhancing network performance and efficiency by considering various effective network parameters [150]. The papers present in this subsection collectively focus on resource optimization in enabled 5G networks, employing ML techniques to enhance network performance.

Some approaches utilize CNN-based models to optimize slice resource allocation and reduce operational costs in 5G networks. Work [151] proposed a CNN scheme for slice management of resources across a variety of case studies in 5G systems, which can predict the capacity needed to support incoming traffic demands within specific network slices while determining the required balance for the operator between resource overprovisioning and SLA violation.

Other studies apply RNN-based solutions to allocate radio resources and avoid traffic congestion dynamically. The authors in [152] propose a new framework that consists of a deep tree model utilizing RNN for the intelligent allocation of radio resources in 5G networks. The objective is to predict the adaptive uplink and downlink ratio in order to avoid future traffic congestion.

In [153], a multitasking learning model using AE, MLP, and RNN for mobile edge traffic classification and prediction, the aim is to start the actions faster and locally by processing the data directly at the edge, exploiting data mining from the control channel. The proposed approach integrates classification and prediction tasks with automatic encoders, providing a unified solution that is adaptable to various system configurations.

A significant number of recent works apply DRL to resource optimization across C-RAN, NG-RAN, and MEC scenarios. In [154], a DRL-based algorithm is proposed to generate policies of self-learning baseband function placement and routing. In two different C-RAN and Next-Generation Radio Access Network (NG-RAN) architectures, the proposed solution aims to minimize the consumed BW and transport latency. In [155], a RAN slicing method is proposed that flexibly allocates RAN resources using distributed DRL, aiming to enhance segment requirement satisfaction and efficient resource utilization. In [156], an FL-based sensing and collaborative learning scheme involving Unmanned Aerial Vehicles (UAV)s for applications in the Internet of Vehicles (IoV) paradigm is considered. In this proposal, a DRL approach may also be adopted to optimize resource allocation.

The paper [157] introduces an energy-efficient resource allocation mechanism for MEC in 5G networks. Utilizing DRL, the authors optimize power consumption and

computational resource distribution across multiple MEC servers.

Expanding on this approach, the work [158] addresses resource optimization in MEC-enabled 5G networks using a DRL-based method. The focus here is on minimizing energy consumption and latency while ensuring efficient task offloading between MEC servers and user devices.

Further exploring resource optimization in MEC, the work [159] proposes a DRL algorithm trained with FL to manage dynamic traffic and optimize resource distribution across edge servers. By focusing on computational offloading and storage management, the approach enhances network efficiency and reduces operational costs. An intelligent dynamic resource allocation framework for resource optimization is introduced in [160], which uses DRL to optimize resource allocation in next-generation wireless networks. The framework dynamically allocates resources between high-bandwidth enhanced Mobile Broadband (eMBB) services and low-latency URLLC by balancing their conflicting needs through resource puncturing and intelligent scheduling. The framework employs two DRL agents, one for eMBB scheduling and another for URLLC puncturing.

Finally, some studies focus on FL to improve resource optimization efficiency while preserving user privacy. In [161], a new approach based on FL with selective aggregation for image classification in vehicular edge computing is presented. The local CNNs models are selected and sent to the central server based on local image quality and computing capacity.

These studies demonstrate how ML can optimize resource allocation and improve overall performance in enabled 5G networks.

4) CONNECTIVITY OPTIMIZATION

Resource allocation is one of the techniques that allows achieving connection efficiency in 5G and beyond networks. These works described below examine advanced resource optimization strategies for 5G and 6G networks, with a focus on enhancing network connection efficiency and spectrum utilization across various service types.

Selected studies apply SL-based approaches to improve communication efficiency. In [162], a novel path planning approach is proposed for a population of UAVs being affected by random wind perturbations in the environment. To this end, the objective is to minimize transmission time, motion energy, and interactions among the UAVs. Here, a SL-based function approximator is used to approximate the solutions of the Hamilton-Jacobi-Bellman (HJB) and Fokker-Planck-Kolmogorov (FPK) equations. An FL approach is then proposed that can share the parameters of the ANN model in drones.

Several contributions utilize RNN-based architectures for traffic prediction and proactive mobility management in 5G and 6G systems. The authors in [163] used an encoder-decoder RNN-based model to predict data traffic incurred at the edge devices of a 6G cloud-native-enabled network. Using time-series data acquired from edge μ-boxes, the model is trained. After that, the model predicts future observations of various data traffic features.

In [164], a new proactive resource management framework is proposed to achieve better handover prediction accuracy for 5G vehicular systems. The module joins RNN with Markov chains to select the next serving BS.

A growing number of studies apply DRL to enhance connectivity optimization by enabling intelligent decision-making for handoff, scheduling, and spectrum management. The authors in [165] addressed optimizing the limited resources at the network edge, aiming to overcome the latency limitations of IoV and smart cities applications. The authors proposed a DRL algorithm utilized in the controller, which in turn decides whether the request has to be served locally by the edge node or allocated to the cloud. To demonstrate the concept of the algorithm, the applications were categorized into several groups with varying latency requirements.

In [166], a channel and latency-aware radio resource allocation algorithm based on DRL is proposed. The algorithm is designed to adapt to channel variations and jointly enhance Bit Error Rate (BER) and latency performance compared to conventional single-rule schemes.

In [167] addresses the challenge of managing radio resources for both eMBB and URLLC in 5G and beyond networks. To efficiently allocate resources, the authors propose a semi-supervised learning and DRL framework. By dividing the resource block allocation for eMBB and URLLC scheduling, the framework ensures low latency and reliability for URLLC while maintaining high throughput for eMBB. Shifting toward future network paradigms, the work [168] proposes a hierarchical DRL-based architecture for resource optimization in 6G networks, focusing on O-RAN and Cell-Free mMIMO systems. The system utilizes multi-agent learning to manage network slices and resources, resulting in enhanced data rates and reduced latency.

A few works leverage RL to enhance handover and mobility prediction. In [169], a user-based interior mobility prediction method is proposed using a Markov chain with an initial transition matrix acquired from RL algorithms in a mmWave heterogeneous network environment. As a second step, based on the acquired knowledge, a preventive handover algorithm is presented to maintain high QoS for indoor users during indoor-to-outdoor transitions.

FL is also used to improve communication reliability and learning accuracy in distributed vehicular systems. In [170], the authors aim to decrease the FL affectation in IoV due to the failure of communication links and missing nodes, especially when continuous exchanges of model parameters are required. For this, they propose using UAVs as wireless relays to facilitate communication between the IoV components and the FL server, thereby improving the accuracy of learning. Specifically, a new joint auction and coalition formation algorithm is proposed to achieve a stable partition of UAV coalitions in which an auction scheme is applied to solve the allocation of UAV coalitions.

DNN-based models have also been proposed for predicting handover success and improving connectivity. In [171], a prediction approach is presented to improve the reliability of the handover success rate and reduce the signaling overhead in the 5G network. This conditional handover approach utilizes DNN and considers sudden changes in the received Reference Signal Received Power (RSRP) power to select the next BS.

Multi-architecture learning frameworks combining RNN, CNN, and DNN are also explored for vehicular SDN-based traffic classification. In [172], these models are tested for resource allocation, where the RNN achieves the best performance with an accuracy of 99.36%.

Lastly, optimization-based methods continue to play a crucial role in enhancing connectivity. In [173], a semi-distributed uplink resource allocation scheme is proposed for user-centric, cell-free MIMO networks, balancing centralized data rates with distributed scalability. Using a Signal to Interference & Noise Ratio (SINR)-based metric and fractional programming, it optimizes user scheduling and power allocation without real-time AP coordination. This hybrid method, which employs a combination of centralized and decentralized control, dynamically adjusts resources to enhance connection optimization and scalability by leveraging fractional programming and compressive sensing.

Together, these studies demonstrate the significant impact of AI and ML frameworks in optimizing resource allocation and improving spectrum efficiency across 5G and 6G networks.

5) COMMENTS AND DISCUSSION

The evaluation of AI techniques in the resource allocation and management subsection provides valuable insights into their diverse applications across EE, latency minimization, resource optimization, and connectivity optimization objectives. Figure 6 highlights the significant role of DRL as the leading AI technique across all categories, reflecting its effectiveness in managing complex and dynamic resource allocation scenarios.

At the group level, tasks are distributed across four key objectives: minimizing latency (29.2%), energy efficiency (27.7%), resource optimization (26.2%), and connectivity optimization (16.9%). This balanced distribution indicates that AI techniques are being applied to address various critical aspects of resource allocation in next-generation networks.

The minimizing latency category has DRL as a dominant AI method with %, followed by DNN and RL at 15.8% each.

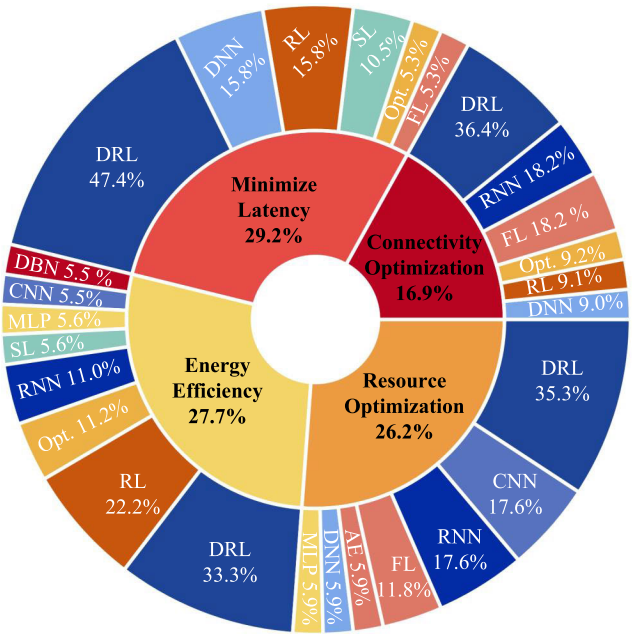


FIGURE 6. Distribution of AI algorithms for resource allocation and management.

SL has the ratio of 10.5%, then there are optimization and FL are equally represented at approximately 5%, showing their specialized applications in reducing latency.

In the domain of energy efficiency, DRL is again the most frequently used method with 33.3%, followed by RL at 22.2%. Optimization and RNN are equally illustrated at around 11%. Techniques such as SL, MLP, CNN, and DBN are employed less frequently, each contributing approximately 5.5%.

The third part in resource allocation and management is resource optimization, where DRL continues to excel, accounting for 35.3%. CNN and RNN share 17.6% each, followed by FL with 11.8% indicating their importance in connectivity-related challenges. Techniques such as AE, DNN, and MLP each account for approximately 5.9%, reflecting their roles in addressing nuanced optimization utilization.

The last part is connectivity optimization, we can notice that DRL leads with 36.4%, followed by RNN and FL, each at 18.2%. Optimization and RL techniques both account for approximately 18%, while DNN contributes 9%. This spread demonstrates a balanced reliance on both advanced and traditional AI techniques for addressing connectivity issues.

Regarding the used tools/libraries to apply the AI/ML method, the following libraries are the most used: TensorFlow (17%), Keras (9.4%), Scikit-learn (7.5%), NumPy (3.8%), and PyTorch (1.9%). Custom codes based on Python (7.5%) were implemented as training methods. In the other 50.9% of the works, no information was provided regarding the used AI/ML tools/libraries.

The tools/platforms used to implement the proposed scenarios in the resource allocation and management section

[6]Not published yet.
[7]Not published yet.
[8]https://data.world/uci/heart-disease
[9]Not published yet.
[10]https://doi.org/10.1109/TSG.2015.2403287
[11]Not published yet.
[12]https://alpercinar.com/open-cell-id/
[13]https://github.com/Whale2021/Dataset/tree/main/didi

TABLE 3. Summary of AI-based resource allocation and management strategies. Category: Energy efficiency.

Ref, Year	Network Domain	Scenario Realization	Scenario validation tool	System Model /Topology	AI Algorithm Type	ML Tool	Dataset Source	Used HW	Run-time
Energy Efficiency									
[129], 2019	5G	Simulation	X	one BS in circle 300m radius 4 UEs, 2 REs, BW= 4 MHz	DBN	TensorFlow	Synthetic	✘	✔
[116], 2019	Fog Vehicular	Simulation	SUMO	fog vehicles interact with roadside units, wireless connectivity successful transmission rate 0.9 or 0.95	RL	X	Synthetic	✘	✘
[121], 2020	C-RAN	Simulation	X	7 RRH, 7 unicast users, 7 multitask users	RNN, DRL	X	Real data [6]	✘	✔
[125], 2020	Mobile Edge Computing, IIoT	Simulation	X	1 BS in 500 m radius, 1 MEC, 10 - 40 IIoT devices	DRL	X	Synthetic	✘	✔
[123], 2020	Edge-Cloud Vehicular	Simulation	X	100 m road, 5 roadside units 1 cloud server, VEC servers 10 vehicles	DRL	TensorFlow	Synthetic	✔	✔
[113], 2021	F-RAN	Simulation	MATLAB	Fog-RAN network, 1 MBS radius 500m, 20 fog-AP change randomly in MBS, 200 Fog-UEs randomly distributed	Optimization	X	Synthetic	✔	✘
[120], 2021	Edge, RAN	Simulation	NS-3	7 BSs have 21 sectors, 10 UEs per sector	RNN	Keras	Simulated	✔	✔
[124], 2021	MEC	Simulation	X	MEC environment, 1 BS, 1 MEC server, 2-30 users	DRL	X	X	✘	✔
[117], 2021	5G, MIMO	Simulation	3D-Ray-Tracing	mMIMO BS (16X8), 40 UE, 7 UE moving with a speed of 1.5m/s	RL	X	X	✘	✔
[126], 2021	Mobile Edge Computing, IoT	Simulation	MATLAB	$100 \times 100m^2$ 2D area, 1 MEC server multiple UEs, random sensor positions	DRL	MATLAB	Synthetic	✔	✔
[122], 2021	Wifi network environment	Simulation	X	APs with different user activities, (RMSE, MAE, R2, Var) performance metrics, in (15, 30, 45, 60 minutes)	CNN	TensorFlow	Real data [7]	✘	✔
[130], 2022	IoT, Edge Cloud	Testbed	Testbed	Cloud node, 2 fog nodes , 1 GW	MLP	Scikit-learn	Real data [8]	✔	✔
[115], 2022	5G RAN	Simulation	X	2 microcell BS, one BS mostly active one BS is active only in public events	SL	X	Real data [9]	✘	✘
[114], 2022	Mobile IoT	Simulation	X	Implement a transmit antenna selection (TAS)-based secrecy scheme employing amplify-and-forward (AF) relaying	Optimization	X	Synthetic	✘	✔
[127], 2024	IIoT, Edge	Simulation	X	3 LTE macro cells, 12 5G NR small cells 63 eMBB users, 99 URLLC users	DRL	X	Simulated	✘	✘
[118], 2024	5G O-RAN	Simulation	X	16 user equipment distributed in a 500m radius Multiple UAVs	RL	X	Simulated	✘	✘
[119], 2024	5G RAN	Simulation	X	Macro BS with service coverage radius at 1,500 m and a small BS at 850 m	RL	X	Simulated	✘	✘

presented some variety: network simulators similar to NS-3 with a 4.1% rate and a rate of 2% for each of the following, iFogsim, ONE, and SUMO. MATLAB occupies 6.1% of the used tools, and custom Python-based code comprises 2% of the proposed scenarios. The Mininet-WiFi emulator is used in 2% of the applied scenarios. On the other hand, a few experiments were validated using a controlled testbed with an 8.2% or based on real experiments with a rate of 2%.

In resource allocation and management, the majority of the studied works, 51%, are based on simulation but without addressing any details. Nonetheless, in this selected sample, 18.4% do not mention any information about how they validate their proposed solutions.

Another essential aspect to consider when analyzing the usage of AI algorithms is their complexity; herein, only 16.7% of the selected articles addressed the complexity of their algorithms, while the majority of the works (83.7%) did not provide details about the algorithms' complexity.

One of the factors studied when applying AI-based solutions is the time required to reach that solution, which consists of training time and execution time. In the resource allocation and management category, only 27% of the studied articles did not consider the consumed time. On the other hand, 73% of the research mentioned the time required to either train the model or validate it. The time spent applying the AI-based solution is directly affected by the hardware used for experimentation; yet, only 73% of those mentioned the time required for the learning process. This is evident in Figure 7, which shows that only 43% of the works provided details about the hardware used.

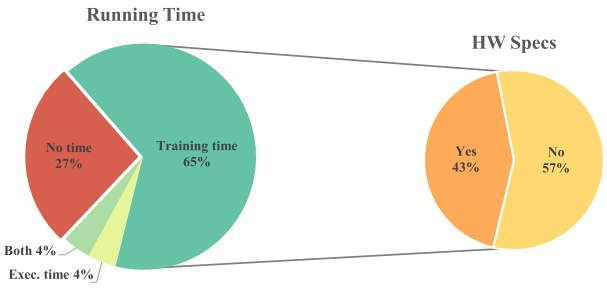


FIGURE 7. Execution time and hardware specifications for resource allocation models.

To provide a more in-depth understanding of the visited articles, Tables 3, 4, and 5 present additional details, including

TABLE 4. Summary of AI-based resource allocation and management strategies. Category: Minimize latency.

Ref, Year	Network Domain	Scenario Realization	Scenario validation tool	System Model /Topology	AI Algorithm Type	ML Tool	Dataset Source	Used HW	Run-time
Minimize Latency									
[140], 2019	Mobile edge computing (MEC)	Simulation	X	The Internet topology zoo [174], MEC with SDN	DRL	Custom code/ Python	Synthetic	✔	✔
[141], 2019	Mobile edge computing (MEC)	Simulation	X	2 Scenarios, 1 centered edge-server in 1000m^2 circle area, 2-12 collaborative edge servers, 10-260 MDs	DRL	X	Synthetic	✘	✔
[138], 2019	Vehicular Fog	Simulation	X	Vehicular fog computing withwithout RSU, 7m between adjacent vehicles	DRL	X	Synthetic	✘	✔
[149], 2019	Vehicular	Simulation	X	Roadside units (RSUs), 100 receiver /transmitter vehicular users	FL	X	Synthetic	✘	✔
[139], 2020	F-RAN	Simulation	X	400x400 m^2, 10 fog-APs, 5 RRHs, and 30 IoT devices	DRL	TensorFlow	X	✔	✔
[135], 2020	Vehicular Fog	Simulation	Custom code/ Python	5 vehicular apps executed in FNs or cloud, data rates 10, 50, 100 Mb/s	RL	TensorFlow	Real data [10]	✔	✔
[146], 2020	Edge	Testbed	Testbed	10 IoT/4 edge devices, 1 edge GW, 120 request/h	DNN	Scikit-learn, NumPy	Simulated	✘	✘
[148], 2020	IoT C-RAN, MIMO	Simulation	X	10 ×10 ×10m room, 128 antennas, 10 IoT devices single-antenna, BBUs linked IoTs to C-RAN	DNN	Keras	Synthetic	✔	✔
[131], 2020	Fog	Simulation	iFogsim	5 different configurations, 1 cloud, 1 fog node, 3 IoT sensors	SL	Custom code/ Python	Simulated	✘	✔
[147], 2021	5G, IoT	Simulation	ONE	10 resource servers, 30 -120 IoT devices, 20 - 200 requests	DNN	X	Synthetic	✘	✔
[132], 2021	HetNet	Simulation	X	Network traffic is classified into 3 classes high, medium, and low priority	SL	Scikit-learn, NumPy	Real data [11]	✘	✔
[134], 2021	HetNet, V-RAN	Testbed	Testbed	1 Edge Host, 2 MTs, each MT linked to 2 USRP B210 boards that implement LTE and IEEE 802.11p	RL	X	Testbed	✔	✔
[142], 2022	Mobile edge computing	Simulation	X	Users 10 - 110	DRL	TensorFlow	Synthetic	✔	✔
[137], 2022	HetNet FRAN	Simulation	X	A layered FRAN environment (MCs, SBSs, MBSs, and a cloud), with cache size varies from 0.1% to 1%.	DRL	X	Synthetic	✘	✘
[143], 2023	IoT, O-RAN	Testbed	Testbed	gNodeBs over a 1 km² area with uniformly distributed eMBB or URLLC end-users.	DRL	X	Real data	✔	✔
[144], 2023	O-RAN 6G	Simulation	Python	20 UEs, 3 RICs each has 5 AI models	DRL	NumPy, TensorFlow	Simulated	✔	✔
[136], 2024	Cloud RAN	Simulation	X	SBSs transmitting at 23 dBm and UEs uniformly distributed in (10 × 10 m)	RL	X	Simulated	✘	✘
[145], 2024	Smart City MEC Federation	Simulation	X	15 Edge Servers, 100 mobile devices, task size: 900-1100 KB, latency tolerance: 600-800 ms	Optimization, DRL	X	Real-world dataset [12], [13]	✔	✘

a brief description of the scenario followed to achieve the specified objectives.

Besides, the tables include information about the source of the datasets used. Analysis revealed that 53% of the datasets used in the papers of this section were synthetic. Moreover, 10.2% of the datasets were generated based on simulation tools, 8.1% of these datasets were created in a controlled testbed, and 16.3% of datasets were chosen from real-world data, like in [115], [121], [122], [130], [132], [151], [153], and [161], Nevertheless 10.2% of the published papers not mentioning the source of their datasets.

C. SPECTRUM MANAGEMENT

Wireless network resource management is a critical issue because traffic demand is growing over time, resulting in a scarcity of available radio resources. As a solid alternative to managing interference, traffic congestion, multi-path usage, multi-channel access, and spectrum management, DL is one of the most promising technologies in 5G wireless network resource allocation [175]. The adoption of intelligent techniques in spectrum sharing management can support the coexistence of heterogeneous radio access technologies and significantly improve capacity and spectrum utilization [176]. The collection of papers presented in this section focuses on enhancing spectral management in 5G and beyond networks, employing advanced techniques like cognitive radios and DL for spectrum management.

1) ENERGY EFFICIENCY

The limited spectrum resources are insufficient to sustain communication systems that demand high data rates. As a result of this trend, spectral efficiency has become the primary performance metric for designing and optimizing wireless systems. However, it also leads to steadily increasing network power consumption, which has serious implications for both economic and ecological costs. As a result, energy conservation has gained widespread recognition as a pressing global issue. The emerging 5G wireless networks

[14] Not published yet.
[15] http://yann.lecun.com/exdb/mnist/, https://btsd.ethz.ch/shareddata/
[16] https://doi.org/10.48550/arXiv.1910.11617
[17] https://openai.com/index/openai-gym-beta/

TABLE 5. Summary of AI-based resource allocation and management strategies. Category: Resource optimization and connectivity optimization.

Ref, Year	Network Domain	Scenario Realization	Scenario validation tool	System Model /Topology	AI Algorithm Type	ML Tool	Dataset Source	Used HW	Run-time
Resource Optimization									
[151], 2019	5G Cognitive Radio	Simulation	X	3 scenarios, varied network DCs capacities in 100 km^2 area, 2 M citizens	CNN	X	Real data [14]	✘	✔
[172], 2020	5G RAN, Vehicular	Emulation	Mininet-WiFi	many V2X communicate with RSUs where V2V and V2I are interacting	RNN, CNN, DNN	Keras	Emulated	✘	✔
[161], 2020	Vehicular edge computing (VEC)	Simulation	MATLAB	Central server, 10 vehicles with velocity (0-15) m/s, 10 types of vehicular clients	FL, CNN	Custom code/ Python	Real data [15]	✘	✔
[153], 2020	Mobile edge computing (MEC)	Simulation	X	eNodeB sends info to connected MTs, 3 groups of apps	AE, MLP, RNN	Keras	Real data [16]	✔	✔
[152], 2020	UDNs	Simulation	X	15 -75 UEs per UDN small cell, UDN small cell has one eNB	RNN	TensorFlow	Synthetic	✔	✘
[155], 2020	5G RAN	Simulation	NS-3	1 BS, 140 UEs, 5 different service types slices, scenarios considering simulation time under NSRS, RBUR, and MRB	DRL	X	Simulated	✔	✔
[156], 2021	HetNet, UAV	Simulation	X	2 scenarios, 5 UAVs/3 subareas, 7 UAVs/6 subareas, final matching based on returned reward and marginal costs	FL	X	Synthetic	✘	✘
[154], 2021	5G, B5G	Simulation	X	Two scenarios, 7 C-RAN nodes, 17 NG-RAN nodes	DRL	X	Synthetic	✔	✔
[157], 2023	C-RAN	Simulation	NS-3	10 fog nodes and cloud servers	DRL	OpenAI Gym [17]	Simulated	✘	✘
[158], 2023	IoT, Edge	Simulation	MATLAB	5-330 tenants, each having 100 users, 500 operator resources, with demand normalized to 0 or 1	DRL	X	Simulated	✘	✘
[159], 2023	5G RAN	Simulation	Python	4 RANs with 10 slices each 800 users in 1200 × 1200 m, pause times of 5-20 seconds	DRL	Keras TensorFlow	Simulated	✘	✘
[160], 2024	5G, 6G	Simulation	Python	Central gNodeB, 12 eMBB users, 6 URLLC users in a 500 m radius area	DRL	PyTorch	Simulated	✘	✘
Connectivity Optimization									
[162], 2020	HetNet, UAV	Simulation	X	25 UAVs at 40m height, travel time 125s step 0.1s, find shortest path to the destination, avoid collision between UAVs	FL	X	Synthetic	✘	✘
[164], 2020	5G, Vehicular	Simulation	SUMO	196 BS with a 40m radius, more than 50 thousands vehicles	RNN	X	Emulated	✘	✔
[171], 2020	5G, B5G mmWave	Simulation	MATLAB	7 mmWave gNBs with 200 m intersite distance, waypoint mobility model generates random UE paths	DNN	Custom code/ Python	Synthetic	✘	✘
[165], 2021	5G, Vehicular	Simulation	X	Edge cluster has 7 FN, 10 utility values of IoV and smart city apps	DRL	X	X	✘	✘
[166], 2021	5G RAN mmWave	Testbed	Testbed	2 UEs using 16QAM, 4 flows, testing SINR per (0 to 800) subcarrier in different scenarios	DRL	X	Testbed	✘	✔
[170], 2021	HetNet UAV, Vehicular	Simulation	X	3 cells, 6 UAVs, allocation UAVs to cells after analyzing the preference and efficiency in the UAV-enabled network	FL	X	Synthetic	✘	✘
[169], 2021	HetNet, mmWave	Simulation	X	3 scenarios, macro BS serving at 3GHz has small cells at 28 GHz	RL	X	Synthetic	✘	✘
[163], 2021	6G	Testbed	Testbed	IoT-Cloud-SDN/NFV enviro., smartX MicroBoxs forming edge clusters	RNN	Keras	Testbed	✔	✔
[167], 2023	5G RAN	Simulation	X	3-cell cluster in 4km area with 22500 users	DRL	X	Simulated	✘	✘
[168], 2024	6G O-RAN	Simulation	Python	Cell-Free mMIMO, O-RAN architecture with distributed DUs	DRL	PyTorch, OpenAI Gym	Simulated	✘	✘
[173], 2024	Cell-Free, MIMO	Simulation	X	7 virtual cells cover 500 m radius, 14, 21, and 28 APs 100 users per km^2	Optimization	X	Simulated	✘	✘

are anticipated to offer improved resource usage efficiency, including EE [177].

DRL is employed to reduce energy use through dynamic spectrum and power allocation. In ultra-dense IoT networks, the authors in [178] proposed a DRL approach that maximizes system throughput and saves energy by utilizing energy harvesting, and determining a power allocation policy for users. The consumed energy, harvested energy, battery-level information of the small BS, and channel-state information are the main parameters of the proposed solution. The authors in [179], propose an architecture for cloud-based training using distributed decision-making DRL scheme is proposed, where radio resource management entities can make lightweight, deep, local decisions while being assisted by on-cloud training and updating.

In [180], an energy-efficient DRL-assisted resource allocation method for RAN slicing in 5G networks is proposed, aiming to improve convergence, reduce complexity, and enhance inter-slice isolation and user acceptance.

DNN with RL improves spectrum use and lowers power in Devise to Devise (D2D) links. In the way to adopt the emerging goals of 5G/6G networks, research [181] presented

a Distributed Artificial Intelligence (DAI) framework combined with ML capabilities, especially DNN with RL to improve spectral efficiency and power consumption in the mobile networks, besides reducing control decision delay. The proposed framework enables serving D2D communication links under different types of spectrum utilization.

RL-based sensing boosts EE in cognitive radio systems. The work [182] introduces an effective spectrum sensing technique for 5G and Wireless Body Area Networks (WBAN) using cognitive radios. By integrating RL, the authors dynamically allocate spectrum resources while minimizing energy consumption. This method significantly improves network EE and extends the lifetime of WBANs, making it particularly suitable for healthcare and IoT applications.

Optimization methods enhance energy-efficient spectrum sharing in MIMO setups. The authors in [183] address energy-efficient resource allocation for connection optimization in a cell-free massive MIMO network utilizing underlay spectrum sharing. The proposed optimization algorithm maximizes EE by optimizing power allocation under interference constraints from a primary network. This approach adapts power distribution across APs to minimize interference with licensed users while maintaining robust connection quality for secondary users.

These studies demonstrate how advanced techniques, including cognitive radios, joint optimization, and reinforcement learning, can enhance spectral efficiency in modern wireless networks, resulting in substantial improvements in performance and sustainability.

2) SPECTRUM EFFICIENCY

Spectrum efficiency is the ratio of data rate to bandwidth and is considered one of the KPIs for wireless networks. Moreover, spectrum efficiency is one of the studied criteria for optimizing wireless networks [184]. These works highlight spectral efficiency optimization techniques for 5G and 6G networks, applying AI models to enhance network performance across various communication scenarios.

SL is used to improve spectrum selection and access in IoT-based sensor networks. The authors in [185] presented an algorithm based on SL to achieve spectrum efficiency by integrating spectrum selection and spectrum access for IoT-based sensor networks. The proposed framework aims to enhance throughput and spectrum access accuracy while reducing access latency in IoT-based networks.

DRL techniques enhance spectral efficiency in power allocation, beamforming, and RAN slicing. The authors in [186] use DRL for power allocation in multi-user wireless cellular networks. The objective is to maximize the downlink cell sum rate in a multi-user wireless cellular network. Simulation results demonstrate that this approach outperforms state-of-the-art model-based methods in terms of sum-rate performance. In [187], a hybrid beamforming design is proposed to improve the spectral, hardware, and computational efficiencies of the system. To achieve this, they formulate the problem of finding the precoding and combining matrices as optimization problems with practical constraints, resulting in a non-convex problem of high computational complexity. To solve this problem, they propose a DRL based solution.

In [188], DRL is used to provide a new intelligent RAN slicing strategy with two-layered control granularity, which aims at maximizing both the long-term QoS of services and Spectral Efficiency (SE) of slices.

In the context of downlink URLLC within 6G networks, the work [189] develops centralized and DRL frameworks to optimize channel access in unlicensed spectrum (NR-U). The study aims to balance energy detection thresholds between NR-U and Wi-Fi systems, reducing interference and collisions.

DNN-based methods boost spectrum efficiency through beamforming, channel prediction, and cognitive radio in diverse 5G scenarios. The authors in [190] proposed a beamforming management framework empowered by DNN for an Intelligent Reflecting Surface (IRS)-assisted mmWave network to handle the beam blockage problem, where the main objective is to reduce system overhead, improve spectral efficiency, and achieve higher handover success rates. DNN was used to train the framework offline to encode the environment geometries, and online learning methods were used for mobility awareness of the device. A new cognitive radio technology with a DNN-based framework in 5G and beyond 5G communication networks was proposed in [191] to achieve better spectrum efficiency. The proposed framework enables a temporary share of the spectrum resources between unlicensed users secondary users (SUs) and licensed primary users (PUs) with mobile primary base stations. In [192], DNN is used to predict the channel gains D2D of apparently independent cellular channels, aiming to improve channel estimation accuracy.

This research [193] proposes DNN solutions for CSI prediction in mMIMO systems to overcome CSI aging caused by environmental dynamics and user mobility. The authors present deep learning architectures based on 3-dimensional complex convolutional neural networks (CCNN) that utilize the angle delay decomposition of past CSI observations to predict future CSI.

A combination of DNN and DRL enables efficient orchestration and spectrum resource allocation in IIoT systems. The authors in [194] use DNNs for real-time inference in end-cloud orchestrated IIoT networks with multidimensional resource constraints. First, in a cloud server in the offline training stage, the parameters of the DNN models are updated via iterative training. In the online inference stage, pre-trained DNN models can be deployed at edge servers or end devices to perform inference tasks. Then, they utilize DRL to establish an end-edge-cloud orchestration architecture that enables the flexible coordination of inference task assignment and DNN model placement, thereby achieving low-latency DNN execution.

GA-based partitioning dynamically balances URLLC and eMBB traffic for spectrum-efficient multiplexing. The authors in [195] explore spectrum efficiency for multiplexing

URLLC and eMBB traffic, using a data-driven genetic algorithm-based spectrum partitioning method. The proposed approach dynamically adjusts spectrum allocation to balance the high reliability and low latency demands of URLLC with the high throughput needs of eMBB. The proposed framework optimizes spectrum efficiency by iteratively refining partitioning based on GA.

Optimization techniques using Reconfigurable Intelligent Surfaces (RIS) phase control and power allocation boost spectrum efficiency in IoT networks. The researchers in [196] investigate spectrum efficiency in a RIS-aided IoT network, focusing on mitigating interference between IoT devices through the optimization of phase shifts of RIS elements. This optimization setup utilizes a conjugate gradient method on Riemannian manifolds for phase optimization and a difference of concave functions approach for transmit power allocation, specifically targeting enhanced spectrum efficiency. The combined approach dynamically adjusts RIS phase shifts to improve spectrum efficiency by directing signals efficiently, resulting in higher spectrum utilization across multiple IoT pairs compared to standard methods.

Together, these studies demonstrate the effectiveness of AL and ML techniques in optimizing spectral efficiency and overall network performance in 5G and 6G environments, offering valuable solutions for various network challenges.

3) SPECTRUM MANAGEMENT OPTIMIZATION

The challenges in the spectrum domain necessitate that 5G networks adopt complex management strategies. Therefore, it is now more crucial than ever to develop methods capable of extracting information from the spectrum in real-time and determining the optimal spectrum access strategy. Thus, real-time, fine-grained spectrum optimization methods will be essential for maximizing the utilization of available spectrum resources for wireless devices [197].

RL-based strategies are applied to optimize delay, energy use, and spectrum access in dynamic wireless environments. A multi-stack RL algorithm was proposed in a mobile edge computing environment by [198]. This algorithm aims to minimize the maximum computational and transmission delay among all users by allocating transmitting power. To improve convergence speed and learning performance, the base station BS in this framework registers historical user information and resource allocation schemes, thereby avoiding the need to learn the same state of user and resource allocation. As a result, the BS has the ability to adjust the uplink/downlink transmit power, as well as the number of subcarriers assigned to each user.

The research paper [199] introduced a RL approach for optimizing energy consumption and enhancing the achievable data rate in Heterogeneous Cloud Radio Access Network (H-CRAN) communication systems for Vehicular Social Networks (VSN). The proposed algorithm employs both centralized Q-learning at the base station and distributed Q-learning in vehicles to enhance the system's EE and simultaneously increase the vehicle's achievable data rate.

In the cognitive railway wireless environment, within the scope of distributed joint spectrum management, work [200] focuses on opportunistic spectrum access and low spectral efficiency in the context of fast motion. The proposed solution presented a multi-agent coordination RL algorithm, which was evaluated by different simulation scenarios.

RNN models improve spectrum sensing and signal classification by learning temporal patterns in user activity. The authors in [201] addressed spectrum management optimization to improve detection performance and classification accuracy in CR networks. The proposed RNN-Sensing Scheme focuses on primary users' activity statistics, such as on and off period duration and duty cycle, while also considering temporal correlation. The experimental validation of the proposed methods utilizes spectrum data from multiple radio technologies, obtained from a designed test-bed measurement setup.

DRL techniques are used for decentralized spectrum allocation, optimizing QoS, latency, and throughput in V2X and smart wireless networks. The authors in [202] propose a new decentralized CSI-independent DRL algorithm based on Multi-Agent Reinforcement Learning (MARL) [203] to maximize the total throughput of Vehicle-to-Infrastructure (V2I) links while meeting latency requirements and reliability of Vehicle to Vehicle (V2V) links in cellular V2X communications. The authors in [204] also adopt DRL for intelligent resource management in smart wireless networks. The goal of the DRL algorithm is to make decisions based on the current status of the network (e.g., channel status, channel quality, device priority, and traffic load) and achieve spectrum sensing accuracy.

The authors of this work [205] present a DRL approach for optimizing URLLC in cyber-physical systems (CPSs) and spectrum allocation in 5G networks. The proposed branching soft actor-critic (BSAC) framework addresses power control, retransmissions, and cognitive radio-based spectrum allocation, improving throughput and reducing interference.

Optimization approaches are employed for efficient indoor network planning and spectrum deployment. Research [206] proposed an indoor GA network planning based on optimization. The proposed model employs a path loss estimation technique to maximize coverage area and minimize path loss. The suggested method focuses on flexible deployment planning with reduced computation time and efficient AP allocation to meet diverse coverage requirements.

DNN-based spectrum sensing improves signal detection and efficiency in low-Signal to Noise Ratio (SNR) 5G CR networks using NOMA. This study [207] proposes an NOMA-based method to enhance cognitive radio networks for 5G IoT applications. It integrates cooperative spectrum sensing (CSS) techniques, Universal Filtered Multi-Carrier, and NOMA to improve spectrum utilization and enhance spectral efficiency. The study employs a DNN model to extract signal characteristics, improving sensing performance in low SNR conditions.

These works demonstrate how AI frameworks can be leveraged to optimize spectrum allocation and enhance throughput in 5G networks, providing practical solutions for both general and IoT-specific applications.

4) MINIMIZE INTERFERENCE

Interference is one of the key challenges in 5G and beyond systems. The interference is a result of the ultra-dense network of small cells (SCs) and the coexistence of several numerologies within the same spectrum [208]. ML and slicing are among the solutions to minimize interference for the 5G and beyond networks [27], [209]. These researchers explore innovative techniques for managing interference in 5G networks, with a focus on optimizing resource allocation and enhancing spectral efficiency in densely populated network environments.

SL techniques reduce interference by enabling efficient detection and predictive modeling in dense wireless settings. The authors in [210] discuss the challenges faced by mMIMO systems in achieving optimal detection at the receiver while minimizing the impact of interference. Although the maximum likelihood detection algorithm can provide the lowest BER, its computational complexity increases as the number of antennas increases. The authors utilize SL to propose an intelligent detection method aimed at reducing computational complexity. The proposed partial learning (PL) detection scheme reduces the impact of interference while reducing computational complexity.

In [211], an SL model is proposed to predict SINR based on the location of a Cyber Physical System (CPS), aiming to reduce Sounding Reference Signals (SRS) transmissions and improve uplink energy use and BW efficiency in 5G networks.

RNN-based models enhance traffic prediction and latency control for interference mitigation in URLLC scenarios. The authors in [212] proposed a traffic-flow prediction algorithm based on RNN with an attention mechanism to train mobile traffic data in single-site mode. The objective is to implement intelligent mobile traffic control, reducing communication latency and packet loss rates in mobile traffic-flow prediction.

DRL approaches optimize resource allocation and reduce interference in V2X, SON, and IIoT systems. A new decentralized resource allocation mechanism for V2V communications based on DRL is presented in [213]. The objective is to respect Vehicle User Equipment (VUE) latency restrictions and reduce interference with other V2I and V2V links. In [214], a DRL-based algorithm for energy-efficient SON architecture based on a tunable eNodeB antenna tilt design for macrocells in a mobile network environment is presented. The proposal aims to improve EE and file transfer time for edge users while reducing handover rejections due to coverage limitations compared to the Stochastic Cellular Learning Automata (SCLA) method.

In [215], the study addresses spectrum management in IIoT networks by minimizing interference through a multi-agent DRL. The primary objective is to manage interference caused by co-channel sharing among clustered IIoT devices while accounting for hardware impairments. A two-step distributed clustering and power management framework is introduced, where devices are first clustered based on SINR using a greedy clustering algorithm. Then, power allocation is optimized via the DRL algorithm. The DRL framework, which does not require precise knowledge of instantaneous signal distortions, adaptively learns and applies power allocation strategies to maximize capacity and reduce interference.

A new model-driven DNN-based block detector was proposed in [216], for Generalized Spatial Modulation (GSM) MIMO inspired by the linear block detectors as Block-Zero-Forcing (ZF) and Block-Minimum Mean Squared Error (MMSE) in order to remove inter-channel interference and noise in practical receivers.

DNN-based solutions improve interference handling in dynamic topologies and signal detection tasks. Work [217] addressed radio resource allocation in an unpredictable and dynamic topology that includes Flying Base Stations (FlyBSs). The authors presented a novel dynamic solution promoted with a DNN algorithm to allocate BW and a transmission power setting that reduces the interference and improves the throughput of the cell edge users, taking into consideration the number of available fly BSs with its' coverage area and the density of the users.

Optimization algorithms dynamically allocate resources to minimize cross-tier interference in dense 5G Heterogenous Networks (HetNets) is proposed in [218], where an Evolutionary Biogeography-based Dynamic Subcarrier Allocation technique is used to address cross-tier interference in 5G HetNets, particularly between macrocells and femtocells. The optimization method is designed to mitigate interference, particularly in dense urban environments and for applications such as connected and autonomous vehicles.

These studies highlight the effectiveness of advanced algorithms in mitigating interference and enhancing spectral efficiency in complex 5G network environments, offering robust solutions for both heterogeneous communication scenarios.

5) COMMENTS AND DISCUSSION

At the group level of analyzing AI techniques in the Spectrum Management subsection, the objectives are distributed across four main categories: spectrum efficiency (31.5%), spectrum management optimization (26.3%), minimizing interference (23.7%), and energy efficiency (18.6%). This distribution reflects the diverse challenges that spectrum management entails and the need for a range of AI techniques to address them effectively. Figure 8 showcases the prominent role of Deep Reinforcement Learning (DRL), which is the leading method across most categories, reflecting its adaptability to dynamic spectrum management tasks.

At the subclass level, in spectrum efficiency, DRL leads with a significant share of 41.7%, highlighting its strength in adaptive learning for dynamic and efficient spectrum allocation. DNN closely follow at 33.3%, showcasing their

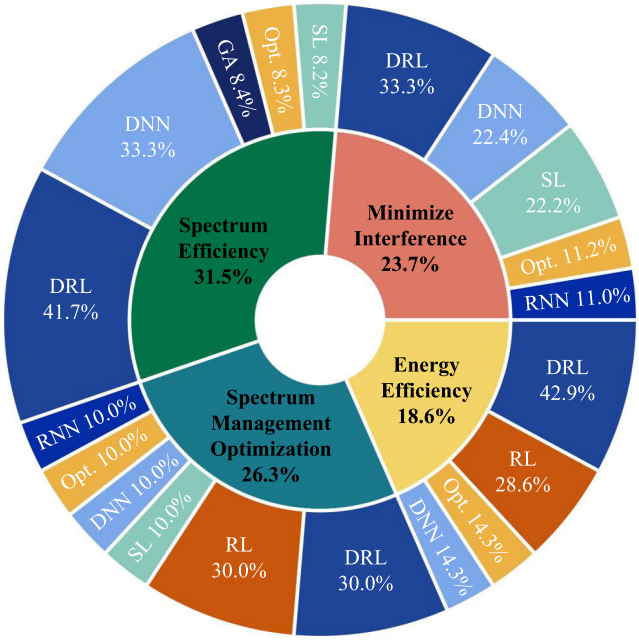


FIGURE 8. Distribution of AI algorithms for spectrum management.

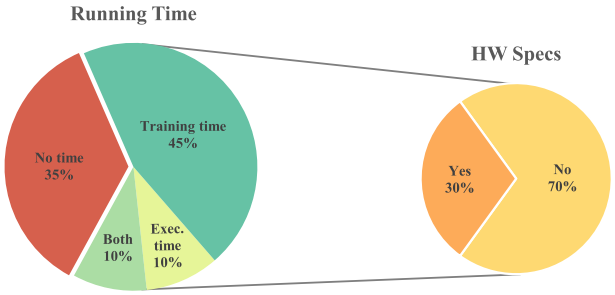


FIGURE 9. Execution time and hardware specifications for spectrum management.

utility in predictive analytics and modeling for spectrum utilization. GA, optimization methods, and SL, collectively account for approximately 25% of the contributions, where each one has around 8% indicating their specialized roles in addressing specific efficiency scenarios.

In the area of spectrum management optimization, DRL and RL are equally represented, with each accounting for 30% of the approaches. SL, DNN, RNN, optimization methods, and RNN each contribute 10%, demonstrating a balanced reliance on both advanced AI methods and more traditional optimization strategies for improving spectrum allocation and usage.

To minimize interference, DRL retains dominance with 33.3%, followed by DNN and SL, both contributing 22.2%. Optimization methods (11.2%) and RNN (11%) complement the analysis, highlighting the importance of predictive modeling and interference mitigation strategies in achieving optimal spectrum performance. The smallest part of spectrum management is EE; herein, DRL again takes the lead with 42.9%, emphasizing its capability to balance energy usage dynamically. RL contributes 28.6%, while optimization methods and DNN share the remaining 28% equally, highlighting a diverse yet focused effort on energy optimization.

The distribution of tools/libraries used to achieve the AI/ML models is as follows: custom code based on MATLAB at the forefront at 20.3%, Keras at 13.8% came in second place, followed by TensorFlow at 10.5%. The least used tools/libraries were PyTorch and Gym at 8.3% each. In the spectrum management section, 46.9% of the papers did not mention the tools/libraries used to validate their model.

Many platforms or simulators are used to validate the proposed scenarios in the spectrum management section. MATLAB held the top spot with 22.2%, while other simulators, namely NS-2 and SUMO, were only utilized in 3.7% of works. Part of the proposed solutions was validated using custom codes written in the R statistical language and C++. However, it should be noted that 48.1% of the works were simulation-based, but without providing details about the experiment. Controlled testbeds were used to validate fewer scenarios, i.e., in 3.6% of the proposed experiments.

Regarding the complexity of AI algorithms in this subcategory, 45.2% of papers clarify the complexity of their algorithms; on the other hand, 54.8% do not provide any information about the complexity. The concept of algorithm complexity gains significance in the context of comparing algorithms within the same domain to gain an overview of existing works.

Another concept used to measure the performance of AI algorithms is the accumulated time to reach the optimal solution. 35% of the scanned research in the spectrum management area did not state the required time to implement their solutions. The training and execution times for the AI algorithm comprise the total training and execution time. The required time to conduct the experiments was specified in 65% of the studies in the selected sample.

As illustrated in Figure 9, this 65% was divided into 45% training time, 10% execution time, and 10% referring to both training and execution times jointly. The hardware used affects the time required for AI algorithms to train. Only 30% of the 65% who identify the needed duration also indicate the hardware utilized in the AI experiment.

Tables 6 and 7 provide further details about the accessed articles, including information on the methods used to generate the datasets. According to the analysis, 80% of the research used synthetic datasets, while 13% of works used datasets generated based on simulation. Testbeds were used to create 3.3% of the datasets.

The predominance of DRL across all spectrum management objectives underscores its versatility in addressing complex and dynamic challenges. The complementary roles of other AI techniques, among others DNN and RL, ensure

[18]WiCa Heuristic Indoor Propagation Prediction Tool, https://www.waves.intec.ugent.be/exposure-tool/expert-edition

[19]SRS: Sounding Reference Signals.

TABLE 6. Summary of AI-based spectrum management solutions. Category: Energy efficiency and spectrum efficiency.

Ref, Year	Network Domain	Scenario Realization	Scenario validation tool	System Model /Topology	AI Algorithm Type	ML Tool	Dataset Source	Used HW	Run-time
Energy Efficiency									
[178], 2020	Ultradense IoT network	Simulation	X	1- 4 mmWave SBSs, each SBS serves UTs, SBSs and UTs have directional mmWave antennas	DRL	TensorFlow	Synthetic	✔	✔
[179], 2020	5G, B5G	X	X	DRL models with different DNN topologies distributed over SONs BS	DRL	X	Synthetic	✘	✔
[180], 2021	5G RAN	Simulation	X	10 slices split into 4 resource-based, 6 rate-based, 200 RAN RB, RB BW 1.8KHz	DRL	Keras, Gym	Synthetic	✘	✔
[181], 2022	5G, 6G	Simulation	Custom code/ Java	Wi-Fi Direct: 255 D2D (200m max), LTE Direct: 2000 D2D (600m max), UE connects to D2D or BS	RL DNN	DL4J	Simulated	✘	✔
[182], 2023	Cognitive Radio 5G	Simulation	X	1- 10 sensor transmission ranges of 5-15 m, data arrival modeled by Poisson distribution	RL	X	Simulated	✘	✘
[183], 2024	Cell-Free, mMIMO	Simulation	X	$250m \times 250m$ CF-mMIMO area, 6 APs, 4 secondary users, $100m \times 100m$ primary network, has 1 base station, 4 primary users	Optimization	X	Simulated	✘	✘
Spectrum Efficiency									
[190], 2020	5G RAN mmWave	Simulation	MATLAB	1 BS, 3 IRSs, many IRS assisted mmWave-vehicles, 1-100 users	DNN	MATLAB	Synthetic	✘	✔
[186], 2020	5G	Simulation	X	25 cells, 1 km spacing, 4 APs per cell, spaced uniformly & randomly in 0.01 km	DRL, DRL	TensorFlow	Synthetic	✔	✔
[192] 2020	5G, Vehicular	Simulation	X	3 scenarios, 250×250m area, 5 BSs, 20 D2D users, 10 cellular users	DNN	X	Synthetic	✘	✘
[188], 2021	5G RAN, 6G network	Simulation	MATLAB	1 slice (uRLLC) higher weighting factor, 1 slice (eMBB), 50 RB, BW 180kHz	DRL, DRL	Keras	Synthetic	✘	✔
[194], 2021	Industrial IoT	Simulation	X	10 Industrial gateways, 35 IIoT devices	DRL	X	Synthetic	✘	✘
[185], 2021	5G, IoT	Simulation	ElectroSense	Different numbers of IoT devices in the range of 15 to 150	SL	Custom code/ Python	Synthetic	✘	✘
[191], 2021	5G , UAV	Simulation	MATLAB	Different symbol modulation schemes, Rayleigh channel	DNN	MATLAB	Synthetic	✔	✔
[187], 2022	mmWave, MIMO	Simulation	Raymobtime	Single user mMIMO systems, antennas, (64/256)Tx antennas, 4 RF chains, multiple data stream	DRL	X	Simulated	✘	✔
[196], 2022	RIS-Aided IoT Network	Simulation	X	$100m \times 100m$ area, 5 IoT device pairs, 64-element RIS at [50m, 100m], and a 10 dBm transmit power cap	Optimization	X	Simulated	✘	✘
[195], 2023	URLLC, eMBB	Simulation	X	200 resource blocks for multiplexing URLLC and eMBB traffic GA initial population: 20-30 individuals	Genetic Algorithm	X	Simulated	✘	✘
[193], 2023	mMIMO 5G, 6G	Simulation	X	1 km radius cell, 260 UEs dropped randomly 60 UEs for testing, 10% for training dataset	DNN	Testbed, Simulation	Testbed, Simulation	✔	✘
[189], 2024	B5G, 6G	Simulation	X	120m x 80m with 3 small cells deployed for WiFi and NR-U, each has 3 gNB, AP nodes, 200 user equipment	DRL	X	Simulated	✘	✘

a holistic approach to spectrum efficiency, interference minimization, and energy optimization. However, the lack of reporting on tools, platforms, and hardware specifications limits the reproducibility and scalability of the proposed solutions, presenting a critical area for improvement in future research.

D. CHANNEL ESTIMATION

In wireless communication systems, the quality of channel estimation has a profoundly critical impact on communication performance and capacity, as the speed and precision of channel estimation directly influence the system. Achieving accurate estimation for the CSI enables maximizing network throughput. In classical channel estimation, the general Expectation Maximization (EM) algorithm is used to estimate the CSI of each multipath component [219], but with the increase in demand on the channels, solutions using more recent AI techniques have emerged [220]. This part presents the attempts of channel estimation in collaboration with AI techniques.

1) ENHANCE CHANNEL ESTIMATION ACCURACY

Coherent detection and precoding in wireless communication systems require accurate estimations of the channel transfer function. In practice, the channel is typically estimated by pilots known to the receiver. These pilots are reused across the cells in a multi-cell mobile network. Pilot contamination happens when a pilot signal is concurrently received at a base station together with other non-orthogonal signals from users in other cells. Thus, it is paramount of importance the accuracy and intuition of channel models, and the models should be based on repeated measurements of how they operate in reality. Integration of AI to the 5G and B5G systems is a promising solution to improve performance and robustness, and adopting DL into the channel estimation methods can enhance the accuracy of channel estimation [5],

TABLE 7. Summary of AI-based spectrum management solutions. Category: Spectrum management optimization and minimize interference.

Ref, Year	Network Domain	Scenario Realization	Scenario validation tool	System Model /Topology	AI Algorithm Type	ML Tool	Dataset Source	Used HW	Run-time
Spectrum Management Optimization									
[200], 2019	Cognitive radio network	Simulation	NS-2, custom code /C++	A railway line, trains avg speed 100ms has 10 chain-distributed cognitive BSs, each CBS has 10 primary users	RL	X	Synthetic	✘	✔
[198], 2020	Mobile edge computing	Simulation	X	MEC network with 100m of radius, 3 BSs/6 users (uniformly distributed), task size 100- 600 kb	RL	X	X	✘	✔
[204], 2020	Edge	Simulation	X	Area with 500m radius, 2500 devices randomly distributed, packet size 2000b	DRL	X	Synthetic	✘	✔
[201], 2020	Cognitive radio (CR)	Testbed	Testbed	1st-USRP-N210 with WBX USRP discone antenna, 2nd-digital spectrum analyzer Rigol DSA-875	RNN	Keras	Synthetic	✔	✔
[199], 2020	H-CRAN	Simulation	SUMO	Single macro cell, 500m intercell distance, 4 RRHS and vehicles light traffic- heavy traffic scenarios	RL	X	Simulated	✘	✔
[202], 2021	5G, Vehicular	Simulation	X	$125 \times 216m$ area, vehicle speed 36 km/s, 4 V2I links, 4 V2V links, BW 4 MHz	DRL	PyTorch	Synthetic	✔	✔
[206], 2022	Indoor 5G	Simulation	WHIPP [18]	Comparing time to compute network deployment and coverage rate, for required APs under different criteria	SL, Optimization	X	Simulated	✘	✔
[205], 2023	5G Edge, Cloud	Simulation	one-G simulation	15×15×11 m³, central gNB, 2 antennas, 5 devices, each with 10 MHz BW	DRL	X	Simulated	✘	✔
[207], 2024	5G, NOMA	Simulation	MATLAB	Secondary users sense the primary users' spectrum and send data to a fusion center	DNN	X	Simulated	✔	✘
Minimize Interference									
[212], 2019	5G Edge, Cloud	Testbed	Testbed	single-site edge cloud, multi-site IoT-cloud, 1 cloud server, 3 local servers, 15 smart devices	RNN	X	Testbed	✘	✘
[213], 2019	5G, Vehicular	Simulation	X	20 -100 vehicles, V2V links 3X number of vehicles, channel BW 1.5MHz	DRL	X	Simulated	✘	✘
[210], 2019	5G, MIMO	Simulation	X	Different scenarios for mMIMO systems	SL	X	Synthetic	✘	✘
[211], 2020	5G, RAN	Simulation	X	The model predicting SINR, then reduces SRS[19] transmission by half	SL	MATLAB	Synthetic	✘	✘
[216], 2020	5G, MIMO	Simulation	X	A generalized spatial Modulation MIMO system, $R_X < T_X$ antennas, QPSK values= 64, 128, 256	DNN	TensorFlow	Synthetic	✔	✔
[217], 2021	Fly BS, B5G	Simulation	X	$1km^2$ area, 400 users, 3 to 24 Flying base stations (FlyBSs)	DNN	X	Synthetic	✘	✘
[214], 2022	5G SON , network	Simulation	MATLAB	Inter-Site distance 0.5-2 Km, BW 5MHz, 45 eNBs, 120 edge sector, 600 mobile	DRL	MATLAB	Synthetic	✘	✘
[218], 2023	5G, HetNets	Simulation	MATLAB	5G macrocells with 30 users, 30 femtocells each with 30 users	Optimization	X	Simulated	✘	✘
[215], 2024	IIoT	Simulation	Python	4 IIoT cells, each with 1 AP 12 devices in 4 clusters	DRL	X	Simulated	✔	✘

[221]. These papers explore advanced channel estimation techniques for 5G systems, employing AI models to address challenges such as signal interference, noise, and multipath fading, with a focus on enhancing the accuracy of channel estimation.

DNN and RNN models are applied to improve mmWave channel estimation and tracking with low overhead. In a vehicular mmWave communications environment, the work [222] addressed the high path loss in mmWave communications. The authors utilize a DNN to achieve better channel estimation and tracking of the mmWave channel using an RNN, where received signals with only an omni-beam pattern at the coordinating BSs are considered to mitigate time overheads for channel estimation.

Work [223] seeks to estimate beamspace channel accurately for 5G mmWave mMIMO by improving DNN-based channel estimation performance. The accurate estimation of the beamspace channel is based on prior information derived from a Gaussian mixture distribution.

The work [224] introduces a hybrid deep learning model that combines convolutional and recurrent DNNs for efficient channel estimation in 5G systems. By leveraging spatial and temporal correlations, this approach effectively addresses issues like multipath interference and signal fading.

In a similar effort, the work [225] focuses on enhancing demodulation reference signal (DMRS) channel estimation in MIMO-OFDM systems using AI-based techniques. The authors investigate DNN models, specifically the CNN-based denoising block and U-Net, to enhance accuracy in low SNR environments. Integrating noise reduction and dimension expansion techniques into the channel estimation process, these models outperform conventional approaches in terms of Normalized Mean Squared Error (NMSE) and BER, marking their suitability for future 5G networks.

Expanding on the use of deep learning for channel estimation, the work [226] proposes an enhanced DNN-based model for 5G MIMO systems. This optimized framework addresses the challenge of signal interference from

multiple transmission paths, significantly improving throughput, SNR, BER, and Mean Squared Error (MSE).

In the context of mMIMO systems in mmWave environments, the paper [227] introduces a DNN model that leverages pilot signals and spatial correlation information to improve channel estimation accuracy for 5G and 6G networks.

A combination of RNN, CNN, and DRL is used to enhance RAN slicing and energy-efficient estimation. The authors in [228] use RNN, CNN, and DRL to RAN slicing for Broadcasting Services to achieve EE.

MLP-based deep models are used to efficiently estimate CSI, enabling improved throughput in low SNR conditions. In [229], the authors propose a deep neural network architecture for the edge of mmWave mMIMO 5G systems, consisting of two MLP layers and a linear output layer. This model generates a high-fidelity channel matrix in a single forward pass between the UE and the mMIMO BS. For the downlink, the BS transmits a reference signal, which the UE uses to estimate the channel and send the information back to the BS for precoding. The goal is to accelerate CSI estimation and support high-throughput communication in challenging SNR environments.

SL is used to extract fading features and reduce signal dimensions, thereby improving CSI estimation accuracy. Estimating the mmWave channel is proposed in [230], where the authors applied SL to extract the channel fading features and reduce the dimension of the received signal, with one-shot training to measure the CSI.

Optimization methods support RIS-aided estimation and activity detection with non-orthogonal pilots. The work [231] addressed activity detection and channel estimation to enhance data transmission in a massive IoT connectivity network aided by a RIS. The work uses an optimization method based on non-orthogonal pilot sequences to solve activity detection and channel estimation issues between the RIS-to-device link and the RIS-to-BS link.

Together, these studies demonstrate the potential of deep learning frameworks in overcoming the limitations of traditional channel estimation methods, offering superior performance in various challenging 5G communication scenarios.

2) IMPROVE THE ROBUSTNESS AND BIT ERROR RATE

The impact of channel estimation on the BER performance under different system parameters to achieve lower BER values [232].

SL improves robustness and detection in OFDM and mMIMO with low-complexity training. The research [233] proposed a channel estimation and equalization scheme for OFDM systems based on SL. The proposed scheme applied one-shot training without a feedback connection between the transmitter and receiver. [234] studied mMIMO systems, where SL is used to provide channel estimation with one-bit Analog-to-Digital Converters (ADCs) feasible, besides achieving better signal detection. The proposed methods are extended to include OFDM systems with frequency-selective fading channels.

DNN reduces noise and improves BER in MIMO and OFDM systems. In order to tackle the non-linear distortions and fading channels resulting from high-power amplifiers, the work [235] used a custom DNN in the MIMO Uncoded Space-Time Labeling Diversity (USTLD) system. The main objective is to minimize channel BW usage based on a blind channel estimator when transmitting power-sharing.

This study [236] presents a deep learning framework focused on BER improvement for massive 5G MIMO-OFDM systems. The authors employ a DNN-based framework, combined with a line sweep-multi-variate kernel density estimation for initial channel estimation and gray code quadratic amplitude modulation for modulation. This approach optimizes both robustness and accuracy. The proposed framework improves BER by reducing noise and interference in high-mobility environments.

GAN enhances joint channel estimation and data detection across space and frequency. In [237], a GAN is used for joint OFDM channel estimation and data detection at different spatial and frequency locations. This work considered a single user with VehA and PedA statistical channel models, where different values of SNR are used to train the network.

CNN boosts estimation accuracy and lowers BER under diverse link conditions. In this study [238], the authors proposed a CNN-based method to address the cascaded channel estimation for 5G and beyond networks. The framework addresses channel estimation under varying quantization settings, link counts, and modulation orders, aiming to improve BER and MSE compared to traditional methods.

DL pipelines refine channel prediction and reduce BER using image-based techniques. Enhancing channel estimation and reducing BER by applying a DL pipeline approach is addressed in [239]. Using adaptive modulation and coding alongside parametric estimation and image processing techniques, the method refines channel prediction for 5G mMTC devices. The DL approach leverages image restoration and super-resolution to clean and upscale noisy, low-resolution channel data, outperforming conventional MMSE estimations, particularly in low SNR conditions. This DL method ensures effective error management and improved BER in challenging channel environments.

Integrating CNN improves joint estimation and BER in noisy OFDM channels. The researchers in [240] propose an attention-guided multi-task CNN to enhance both robustness and BER for joint carrier frequency offset and channel estimation in OFDM systems. By coupling these tasks within a shared branch that extracts common features, the model applies dedicated branches for carrier frequency offset and channel estimation. An attention-guided module minimizes the impact of carrier frequency offset error on channel accuracy, utilizing tailored weights to enhance BER.

Together, these studies demonstrate how AI and hybrid estimation techniques can significantly enhance BER performance and channel estimation accuracy in modern and future 5G networks.

TABLE 8. Summary of AI-based channel estimation methods.

Ref, Year	Network Domain	Scenario Realization	Scenario validation tool	System Model /Topology	AI Algorithm Type	ML Tool	Dataset Source	Used HW	Run-time
Enhance Channel Estimation Accuracy									
[222], 2020	mmWave, MIMO, Vehicular	Simulation	MATLAB	MIMO mmWave, many BSs, each BS has 32 antennas serving a number of single antenna vehicular UEs	DNN, RNN	Keras	Simulated	✘	✔
[223], 2020	5G network, mmWave, MIMO	Simulation	X	mMIMO mmWave, 3 BSs, BS has 256 antennas with 16 RF chains serving 16 single-antenna UEs	DNN	X	Simulated	✘	✘
[228], 2021	5G RAN	Simulation	X	Cell radius 500m, 1st slice with 30 UEs 2nd slice with 70 UEs, and SNR varies from -10 dB to 5 dB	RNN, CNN DRL	X	Synthetic	✘	✘
[229], 2021	B5G, MIMO	Simulation	MATLAB	Symbol modulation QPSK with OFDM, downlink system with 32×4 MIMO mmWave channel	MLP	MATLAB	Synthetic	✔	✔
[230], 2021	mmWave	Simulation	X	Symbol modulation 16QAM, OFDM mmWave-based system	SL	X	Synthetic	✘	✘
[231], 2021	IoT	Simulation	X	An uplink system, BS has 40-60 receive antennas serve 1000 single antenna-IoT devices, 80 of them are active	Optimization	X	Synthetic	✘	✘
[224], 2023	5G, MIMO	Simulation	MATLAB	Symbol modulation 16-QAM, 2×2 MIMO, TDL-C channel model	DNN	X	Synthetic	✘	✘
[225], 2023	5G OFDM-based MIMO	Simulation	X	32-antenna MIMO, 16-QAM modulation, mixed CDL (A-E) channel models	DNN	X	Synthetic	✘	✘
[226], 2023	5G, MIMO	Simulation	NS2	QPSK modulation, MIMO with AWGN channel	DNN	X	Synthetic	✘	✘
[227], 2024	5G, mMIMO	Simulation	X	MIMO 8-16 antennas, 80 GHz frequency, 32 subcarriers, 32 channel paths	DNN	X	Simulated	✔	✘
Improve the robustness and Bit Error Rate (BER)									
[233], 2019	OFDM	Simulation	X	M-QAM symbol modulation, OFDM system under different multi-path fading channels	SL	X	Synthetic	✘	✘
[235], 2020	5G network, MIMO	Simulation	X	Symbol modulation 16-QAM and 16-PSK, 2×4 MIMO with Rayleigh fading channel	DNN	X	Synthetic	✘	✘
[237], 2020	OFDM	Simulation	X	Symbol modulation 256 QAM with a single antenna OFDM, 2 statistical channel models	GAN	PyTorch	Synthetic	✔	✘
[234], 2021	mMIMO, OFDM	Simulation	X	OFDM mMIMO, symbol modulation QPSK and 16QAM, BS has multiple antennas and serves many single-antenna UEs	SL	Scikit -learn	Synthetic	✘	✔
[238], 2022	5G network	Testbed	Testbed	Considering BER as a performance metric under non-identical fading channels	CNN	X	Testbed	✔	✔
[239], 2022	5G, mMTC	Simulation	X	127 pilot subcarriers, 8-time slots per frame for channel estimation	DL	X	Simulated	✘	✘
[240], 2023	OFDM, 5G/B5G	Simulation	Python	EVA channel model, 16 QAM modulation, 600 subcarrier, 10 Mhz bandwidth	CNN	Pytorch	Simulated	✔	✘
[236], 2024	mMIMO, OFDM	Simulation	MATLAB	MIMO with Rayleigh fading channel, 2 to 8 Tx antennas, 2 to 8 Rx antennas	DNN	X	Simulated	✘	✘
Improve Spectral Efficiency									
[241], 2020	mmWave, MIMO	Simulation	X	one BS has 64 R_X, 4 Rf chains, 3 UT with single antenna	CNN	TensorFlow	Synthetic	✘	✘
[245], 2021	5G , OFDM	Simulation	X	3 scenarios, symbol modulation QPSK, SISO OFDM system, slow fading channel	SL	X	Synthetic	✘	✘
[248], 2021	5G , OFDM	Simulation	X	one BS, downlink multi-user SISO OFDM-NOMA based system under Rayleigh fading channel	RNN	X	Synthetic	✘	✘
[242], 2021	B5G, MIMO	Simulation	X	mmWave MIMO AP has 36 transmit antennas serving single antenna UE	CNN	X	Synthetic	✘	✔
[243], 2022	mmWave, mMIMO	Simulation	MATLAB	Tx=128, Rx=16, 4 RF chains per side 16 subcarriers at 60 GHz, 2 GHz BW	CNN	X	Simulated	✔	✘
[244], 2022	mMIMO	Simulation	Google Colab	1 km^6 area has 16 cells, each has 10 UEs, base station arrays of up to 256 antennas	GAN	Google Colab	Real data [20]	✘	✔
[247], 2024	Fog-RAN	Simulation	X	F-RAN has, 10 small-cells, 5 F-APs, 50 devices, and 30 dBm max power	DRL	X	Synthetic	✘	✘
[227], 2024	5G, mMIMO	Simulation	MATLAB	8-16 antennas, 80 GHz frequency, 32 subcarriers, and 32 channel paths	DNN	MATLAB	Simulated	✔	✘
[246], 2024	5G, UC-RAN	Simulation	MATLAB	2-4 users, 16-128 RRH antennas, QPSK modulation, 4-cluster mmWave channels	SL	X	Synthetic	✘	✘

3) IMPROVE SPECTRAL EFFICIENCY

Spectral efficiency can be enhanced using channel estimation methods and especially by applying DL techniques [241]. These papers explore advanced strategies to enhance spectral efficiency in 5G and beyond networks, employing deep learning and approaches to optimize power control, channel estimation, and beamforming techniques.

CNN enhances spectral efficiency via better channel estimation and beamforming. The authors in [241] proposed a channel estimation scheme based on CNN to enhance

the spectrum efficiency for mmWave mMIMO multi-user systems. Beamspace channel amplitude estimation and channel reconstruction form the presented scheme, where the beamspace channel's dominating entry indices are then acquired, allowing for the reconstruction of the channel.

In [242], a method combining offset learning with CNN is proposed to estimate indoor channels with affordable piloting overhead. This method can dynamically track CSI without requiring any prior knowledge of the IRS-assisted channel structure or internal statistics.

The researchers in [243] introduce DL architectures using CNN for channel estimation and hybrid beamforming in multi-carrier mmWave mMIMO systems. The authors focus on improving spectral efficiency. The proposed frameworks employ CNN to estimate channels from pilot signals and optimize hybrid beamforming, reducing the number of required pilot signals and computational costs.

GAN maps uplink to downlink CSI for efficient Frequency Division Duplex (FDD) estimation. Reference [244] presents a CGAN-based approach for improving spectral efficiency in channel estimation within FDD mMIMO systems. It highlights the limitations of traditional DNN-based methods, which often require large datasets and are susceptible to environmental variations. By mapping uplink to downlink CSI using CGAN, the method achieves higher spectral efficiency due to enhanced estimation accuracy without the full overhead of downlink pilot transmission.

SL improves channel estimation and spectral efficiency in OFDM and C-RAN. Another paper [245] addressed the channel estimation issue in an OFDM system. The authors applied SL in three main scenarios, two of which involved perfect synchronization and one with imperfect synchronization. Herein, the studied OFDM symbol is divided into multiple groups, and the channel estimation process is performed on each group separately. The generated bitstream by the OFDM technique is modulated and then sent using a single-antenna Single Input Single Output (SISO) system through a slow fading channel. The estimator training process was conducted online periodically, using data generated during transmission.

In the context of user-centric C-RAN, the work [246] proposes a SL-model based on a variational Bayesian approach to enhance spectral efficiency and address pilot contamination. The iterative variational Bayesian inference (IVBI) algorithm, initialized with least squares estimation and followed by iterative refinement, demonstrates substantial improvements in NMSE, BER, and spectral efficiency compared to conventional methods.

DRL boosts spectral efficiency through adaptive power control in F-RAN. The work [247] presents a spectral efficiency improvement strategy for downlink F-RAN using DRL. By incorporating ensemble learning, the system dynamically adjusts power allocation across fog access points (F-APs) to counteract interference and hardware impairments.

DNN increases spectral efficiency with accurate estimation and low-complexity beamforming. Focusing on mMIMO 5G systems, the work [227] introduces a DNN-based hybrid beamformer (DLHB-MIMO5G) designed to improve spectral efficiency and reduce computational complexity. Using the algorithm for channel estimation, the system improves accuracy by incorporating orthogonality and user feedback.

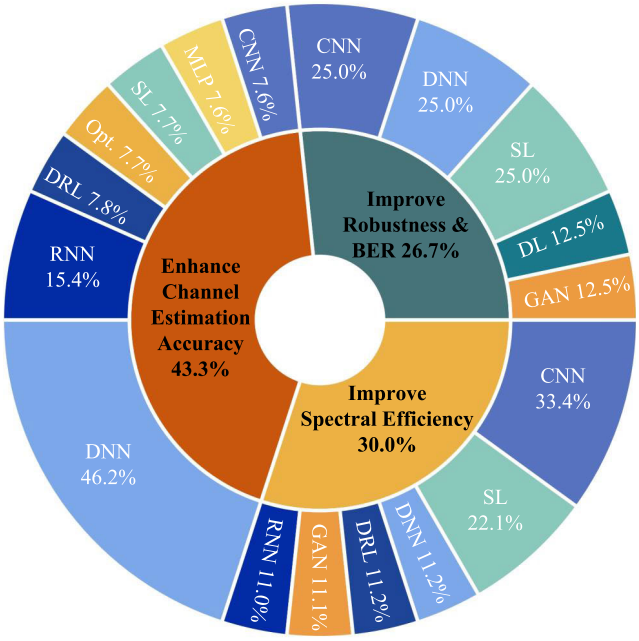


FIGURE 10. Distribution of AI algorithms for channel estimation tasks.

Together, these studies highlight how deep learning and Bayesian techniques can significantly enhance spectral efficiency, optimize power control, and improve channel estimation accuracy in modern 5G network deployments.

4) COMMENTS AND DISCUSSION

Channel estimation is a cornerstone of modern wireless communication systems. The application of AI techniques has significantly enhanced channel estimation, allowing for improved accuracy, robustness, and spectral efficiency. The analysis focuses on three primary objectives: enhancing channel estimation accuracy (43.3%), improving spectral efficiency (30%), and improving robustness and BER (26.7%). As illustrated in Figure 10, the DNN stands out as the most frequently employed method, demonstrating its effectiveness in managing the complex nature of channel estimation problems. Within the category of enhancing channel estimation accuracy, DNN leads with 46.2%, underlining its strength in capturing intricate data relationships. RNN hold a notable 15.4%, emphasizing their suitability for sequential data processing. Other techniques, in particular DRL, optimization methods SL, MLP, and CNN contribute around 7.8% each, showcasing their adaptive learning capabilities.

For improving spectral efficiency, CNN leads with 33.4%, leveraging its ability to extract meaningful features from high-dimensional data, followed by SL at 22.1%. DNN

[20]https://renew.rice.edu/dataset-fdd.html

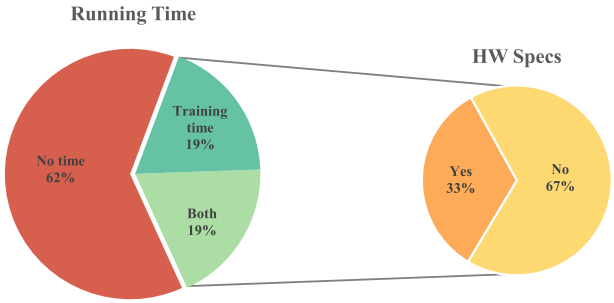


FIGURE 11. Execution time and hardware specifications for channel estimation models.

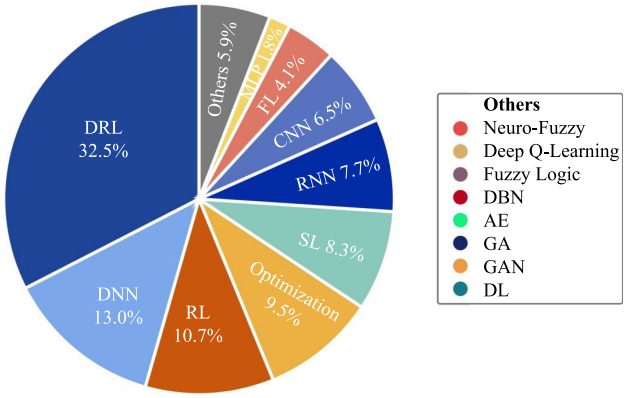


FIGURE 12. Overview of AI algorithms' distribution across 5G and beyond network domains.

and DRL each account for 11.2%, while GAN and RNN contribute 11.1%, highlighting their application in creating robust channel models. In improving robustness and BER, CNN, DNN, and SL equally dominate at 25%, demonstrating their collective strength in addressing noise, interference, and reliability challenges. DL and GAN each add 12.5%, showing their utility in enhancing channel robustness.

The tools/libraries used to verify the proposed AI model regarding the studied papers in this section were MATLAB, TensorFlow, Keras, PyTorch, and Scikit-learn, with each accounting for approximately 6%. The 68.8% do not mention the tools/libraries used to validate their AI models. Similarly, regarding the platforms used to confirm the proposed solutions, MATLAB was the only simulator mentioned among the studied samples, with an 18.8% rate. Moreover, 6.3% of the experiments were implemented through a controlled testbed. On the other hand, 75% of the studies did not mention the platform or the framework used to achieve their proposed scenarios.

An algorithm's estimated computing power requirement can be determined by its time complexity. The importance of this concept stems from its ability to predict the amount of memory and time required to execute an algorithm. Only 62.5% of these articles state the complexity of the AI methods, while the other 37.5% omit this information. Figure 11 demonstrates that most articles (62%) did not mention the algorithms' running time. While just 38% of the publications referred to the running time when discussing the AI algorithms' running times. 19% of the publications provided both training time and execution time, while the other 19% referred only to the training time. In addition, only 33% of this 38% confirmed that the hardware used features that specifically influence training and execution times. Table 8 describes the experiment's scenario and further information about the publications that were looked at. Additional details about the different datasets used are also included in the table. According to the analysis, 81.25% of the datasets used in this section's research were created synthetically, 12.5% were generated based on simulations, and the remaining 6.25% of datasets in this area were generated in a testbed.

The dominance of DNN across all objectives reinforces its versatility and reliability in channel estimation tasks. Meanwhile, the contributions of CNN, RNN, GAN, and SL highlight the diversity of approaches tailored to address specific challenges, such as accuracy, spectral efficiency, and robustness. However, gaps in reporting tools, simulation platforms, and hardware specifications remain a persistent concern, underscoring the importance of transparency and standardization in future research.

V. FINAL COMMENTS AND DISCUSSION

A. AI ALGORITHMS DISTRIBUTION

The analysis of AI algorithms applied across various tasks in 5G and beyond networks highlights a diverse and dynamic distribution of techniques, as depicted in Figure 12. Among all algorithms, DRL emerges as the most dominant, accounting for 32.5% of the methods used. This highlights its versatility and effectiveness in solving complex, real-time decision-making problems, such as resource allocation, task offloading, and spectrum optimization.

DNN holds the second position with 13%, showcasing its reliability in handling high-dimensional data for tasks such as channel estimation and improving spectral efficiency. RL follows with 10.7%, reflecting its growing role in autonomous and adaptive learning-based scenarios, particularly in dynamic resource management. Optimization techniques contribute 9.5%, underscoring their utility in fine-tuning performance metrics such as EE, latency minimization, and load balancing.

In the mid-range, SL occupies 8.3%, indicating its relevance in scenarios where labeled data is available for predictive tasks, e.g., traffic prediction and interference minimization. RNN account for 7.7%, showcasing their strength in processing sequential and time-series data, particularly for tasks involving mobility and traffic prediction. CNN contributes 6.5%, where it excels at feature extraction and dimensionality reduction for tasks such as spectrum management and channel estimation.

The Others category collectively accounts for 5.9% and includes methods as Neuro-Fuzzy Systems, Deep Q-Learning, Fuzzy Logic, and advanced models like DBN, AE, GA, and GAN. These methods, although less frequent,

offer specialized solutions for unique challenges, such as EE, robustness, and model generation, in complex network environments. Notably, FL represents 4.1% of the distribution, demonstrating its emerging importance in decentralized AI frameworks that prioritize privacy and efficiency.

The dominance of DRL, DNN, and RL highlights their pivotal roles in addressing critical challenges in 5G and beyond networks. While optimization and classical ML techniques remain significant, emerging methods like GAN and FL show promise for future advancements.

In emerging RAN architectures, particularly within AI-native and distributed frameworks, it is common to deploy multiple AI/ML algorithms in parallel, each targeting optimization of critical KPIs such as latency, energy, or throughput. However, without coordinated decision-making, these agents may conflict by issuing contradictory actions or overfitting to partial objectives. This is especially critical when agents operate asynchronously or across different network segments. Best practices for conflict management involve designing centralized arbitration modules, policy-aligned reward functions, and MARL mechanisms that promote cooperative behavior. Lifecycle orchestration frameworks, e.g., 3GPP, Network Data Analytics Function (NWDAF), and European Telecommunications Standards Institute (ETSI), are also recommended for monitoring and regulating overlapping model actions. Despite recent progress, the absence of standardized conflict resolution protocols remains a significant barrier to the scalable and safe integration of AI in RAN systems.

B. ML TOOLS/LIBRARIES

The process of training, testing, and validating ML models is a fundamental component of AI-based research, particularly in wireless communication systems. A critical analysis of the reviewed articles highlights a persistent issue: a significant portion of studies (59.7%) do not specify the tools or libraries used for implementing and training the ML algorithms, as shown in Figure 13. This lack of transparency hinders reproducibility and undermines confidence in the reported results, posing challenges for future researchers attempting to validate or extend these findings.

Among the studies that do report the tools used, TensorFlow emerges as the most frequently utilized library, representing 10.4% of the total. TensorFlow's popularity is attributed to its scalability, versatility, and strong support for deep learning models. Customized ML tools, often tailored for specific tasks using Python or MATLAB, account for 9.1%, reflecting the flexibility of coding environments in meeting unique research requirements. Keras is employed in 7.1% of the studies, particularly for prototyping and developing deep learning models. PyTorch, a leading framework for research-oriented deep learning tasks, contributes 5.2% of the reported tools, highlighting its growing adoption due to dynamic computation graphs and strong community support. Scikit-learn, a popular library for classical machine learning algorithms, accounts for 3.2% of the studies, demonstrating its relevance in scenarios that require traditional supervised and unsupervised learning methods. The Others category comprises 5.2% and includes tools such as NumPy, OpenAI Gym, DL4J, Gym, and Google Colab, each contributing to specific stages of ML development, testing, or deployment.

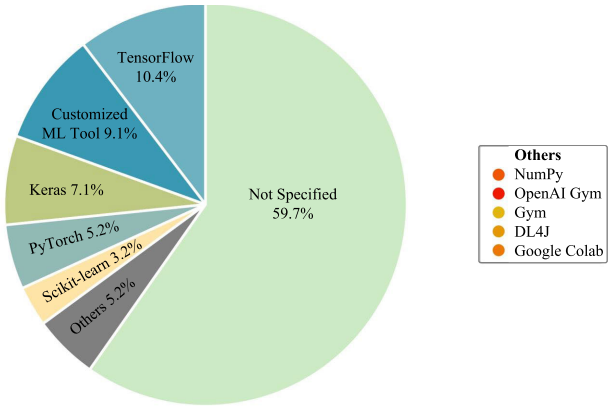


FIGURE 13. Comparative analysis of machine learning tools and libraries employed in studies.

It is important to note that the choice of library or tool used for training the ML model can significantly impact the model's performance and the results obtained. As such, sufficient detail must be provided in the study, including the tool or library used, to ensure that the results can be accurately reproduced. This can enhance the credibility of the study and increase confidence in the results obtained.

C. DATASETS GENERATING SOURCE

The analysis of datasets used in AI-based studies reveals a notable reliance on artificially generated data, as shown in Figure 14. Specifically, 47% of the reviewed articles utilized synthetic datasets, which are artificially created and not derived from real-world environments. This highlights the challenges of obtaining real-world data while showcasing the flexibility that synthetic datasets offer for controlled experiments. However, their limited ability to replicate real-world complexity raises concerns about the generalizability of results.

Simulated datasets account for 32.9%, making them the second most commonly used source. These datasets are generated using simulation tools, among others, MATLAB with a ratio of 25%, followed by the network simulator (NS-3 NS-2) with 10.7%. Additionally, tools like CloudSim 10.7% and iFogSim 7.1% provide valuable datasets on the performance and resource utilization of these emerging computing paradigms. The use of other simulators, such as SUMO, which provides controlled environments for validating AI models in network and vehicular scenarios, Raymobtime, WHIPP, and One-G, contributes smaller shares (3.6%), showcasing their utility in diverse application domains. Finally, 32.1% of these datasets are generated based on customized user-defined simulation platforms.

Datasets derived from real-world data represent 10.1% of the studies, indicating an effort to validate AI models in practical environments. These datasets are typically collected

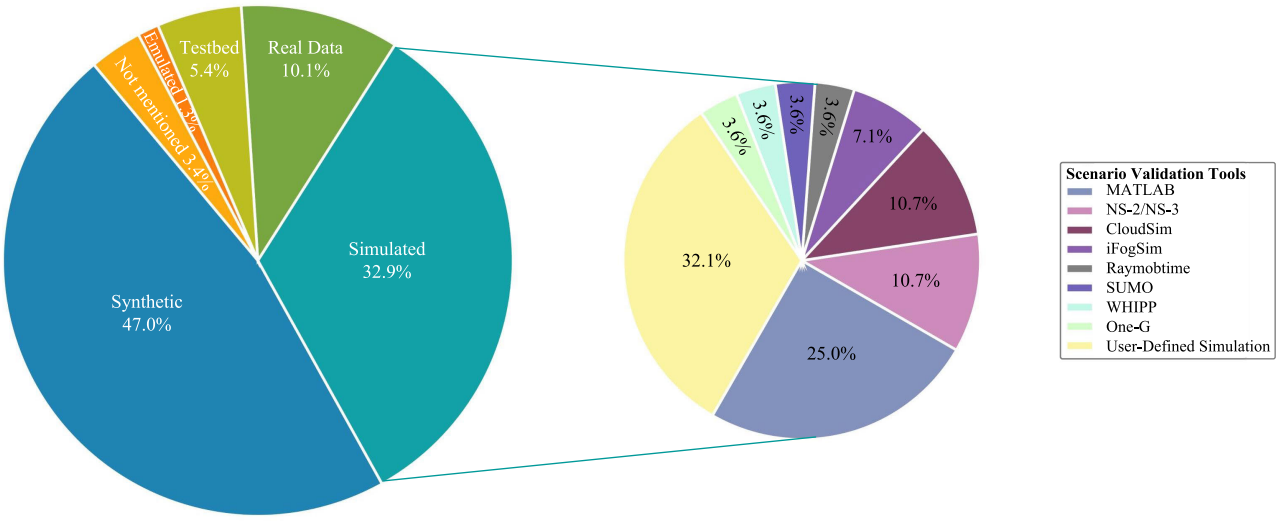


FIGURE 14. Sources of datasets used in AI applications: synthetic, simulated, and real-world.

from production systems, field experiments, or publicly available repositories. However, their lower representation compared to synthetic and simulated sources highlights the difficulty of accessing and processing real-world data, especially in large-scale network experiments.

Testbeds, which account for 5.4%, offer controlled experimental setups that bridge the gap between simulations and real-world environments. They provide valuable validation for AI models but remain underutilized due to their cost and operational complexity. Meanwhile, a small proportion of studies 1.3% relied on emulators like Mininet-wifi to generate datasets, offering lightweight and flexible alternatives for specific network scenarios. Notably, 3.4% of the articles did not mention the source of their datasets, raising concerns about reproducibility and transparency.

It is worth noting that the use of synthetic datasets may limit the generalizability of the results obtained from the articles, as they may not accurately reflect real-world scenarios. Additionally, the lack of details provided about the source of some datasets may make it difficult for others to understand the data and replicate the experiments. On the other hand, the use of real-world datasets, testbeds, and simulators, such as SUMO and NS-3, can increase the credibility of the results and make them more applicable to real-world scenarios.

D. SCENARIO REALIZATION

The evaluation of scenario realization methods highlights a strong reliance on simulations for validating AI-based solutions, as presented in Figure 15. A significant 91% of the studies utilized simulation-based approaches, underscoring their practicality for creating controlled environments that mimic real-world conditions. Simulations offer flexibility, scalability, and ease of experimentation, making them the preferred choice for validating AI models in wireless communication systems.

Among the 91% of simulation tools, MATLAB is the most commonly used, representing 14.1% of the studies. Its versatility and comprehensive libraries make it a valuable platform for scenario modeling and validation. NS-2/NS-3 contributes 5.1%, reflecting its strength in simulating network-based experiments and traffic flows. Additionally, SUMO, CloudSim, and iFogSim are each reported in 2.3% of the articles, particularly for mobility-based and edge computing scenarios, respectively. Custom or user-defined simulation environments, built with Python, Java, or C++, account for 8.8%, highlighting the flexibility of tailor-made solutions for specific research objectives. Other tools, including NetworkX, 3D Ray-Tracing, ElectroSense, Raymobtime, WHIPP, and Google Colab, contribute smaller shares of 5.9%, reflecting their niche applications in targeted experiments. However, a significant portion of articles (59.4%) did not specify the scenario validation method used, which raises concerns about the transparency and reproducibility of these studies. The lack of clarity makes it challenging to evaluate the reliability of the reported findings and to compare results across studies.

Testbeds are employed in 7.1% of the studies, providing physical experimental setups that bridge the gap between simulations and real-world environments. Testbeds offer higher realism but are underutilized due to their cost, complexity, and operational constraints. Emulation techniques, although limited to 1.3%, offer lightweight alternatives for validating AI solutions in controlled environments, often utilizing platforms such as Mininet and Mininet-WiFi.

The dominance of simulations, while practical, emphasizes the need for greater integration of testbeds and real-world validations to ensure AI-based solutions can perform effectively in real-world environments. Additionally, researchers

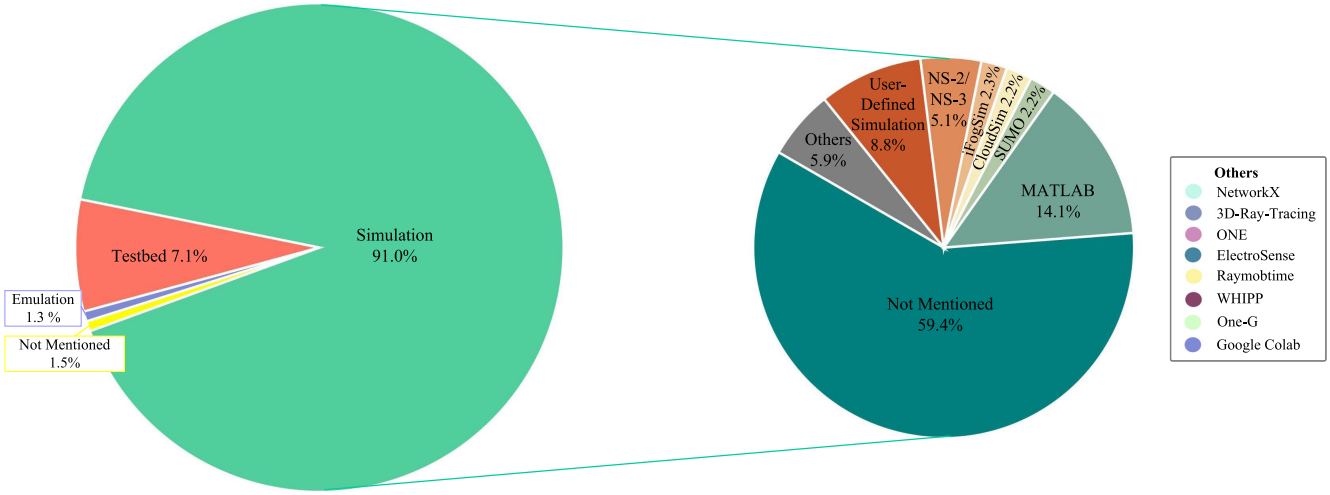


FIGURE 15. Methods for scenario realization: simulation, testbeds, and real-world environments.

should prioritize reporting clear and detailed information about their scenario validation methods, as this enhances the transparency, comparability, and credibility of their findings.

VI. CONCLUSION

This survey provides a comprehensive review of AI and ML applications in RAN for 5G and beyond networks, addressing both achievements and persistent challenges in this growing field.

- The integration of AI algorithms, particularly deep learning approaches including DRL and DNN, has proven highly effective in solving complex tasks like resource allocation, task offloading, and spectrum management. These methods showcase their ability to manage dynamic, real-time decisions and high-dimensional data effectively. However, despite their dominance, many studies fail to provide details on computational complexity, which limits our understanding of their scalability and practical feasibility in real-world applications.
- While synthetic and simulated datasets dominate as sources for AI training and testing, their flexibility has allowed researchers to perform large-scale experiments under controlled conditions. However, the limited use of real-world data and testbed validations raises concerns about the generalizability of findings. Greater adoption of real-world datasets and testbeds would enhance the credibility and applicability of proposed AI solutions in real-world environments.
- The tools and libraries used for implementing ML models, such as TensorFlow, PyTorch, and Keras, offer robust capabilities for model development and validation. Customized tools also demonstrate the adaptability of researchers to unique scenarios. On the negative side, a significant lack of transparency regarding the choice of libraries and tools persists in many studies, hindering reproducibility and comparability across research efforts.
- Scenario realization through simulations has been instrumental in enabling scalable and flexible experimentation, with platforms like MATLAB and NS-3 providing essential environments for AI validation. However, the heavy reliance on simulations alone, coupled with the underutilization of testbeds and real-world emulation, limits the practical validation of AI-driven solutions. Additionally, many studies fail to disclose the specific platforms or methods used, which creates challenges in assessing the robustness of the results.
- Finally, the reporting of computational costs, including training time, execution time, and hardware specifications, remains inconsistent across studies. While some studies highlight these aspects, a large portion still omits such critical details. This omission makes it difficult to evaluate the EE, scalability, and feasibility of AI algorithms for practical deployment in 5G and beyond networks.

In conclusion, the integration of AI and ML technologies has demonstrated significant potential for addressing key challenges in 5G RAN, particularly in resource optimization, spectrum efficiency, and latency reduction. The versatility of deep learning methods, the adaptability of simulation tools, and the flexibility of synthetic data are major strengths. However, this potential is undermined by gaps in reporting computational complexity, dataset sources, tool usage, and validation methods for scenarios. Moving forward, it is essential for researchers to focus on improving transparency, adopting real-world validations, and providing detailed documentation of tools and metrics. By addressing these challenges, the field can progress toward scalable, reproducible, and practically deployable AI solutions for next-generation wireless networks.

TABLE 9. List of used acronyms.

Abbreviation		Abbreviation		Abbreviation	
AC	Actor-Critic	FIS	Fuzzy Inference System	ML	Machine Learning
ACO	Ant-Colony Optimization	FL	Federated Learning	MLP	Multilayer Perceptron
ADC	Analog-to-Digital Converter	FlyBSs	Flying Base Stations	mmWave	millimeter Wave
ADMM	Alternating Direction Method of Multipliers	FRL	Federated Reinforcement Learning	MMAE	Mean Absolute Error
AE	Auto-Encoder	FSO	Free-Space Optical	MSE	Mean Squared Error
AI	Artificial Intelligence	GA	Genetic Algorithm	MT	Mobile Terminal
ANN	Artificial Neural Network	GAN	Generative Adversarial Network	MTC	massive Machine Type Communication
AP	Access Point	GPU	Graphics Processing Unit	NG-RAN	Next-Generation Radio Access Network
API	Application Programming Interfaces	GRU	Gated Recurrent Unit	NGMN	Next Generation Mobile Network
APC	Available Processing Capacity	GSM	Generalized Spatial Modulation	NLP	Natural Language Processing
AR	Augmented Reality	H-CRAN	Heterogeneous Cloud Radio Access Network	NMSE	Normalized Mean Squared Error
B5G	Beyond 5G	HD	High Definition	NN	Neural Networks
BBUs	Baseband Units	HUDNs	Heterogeneous Ultra-Dense Networks	NR	New Radio
BER	Bit Error Rate	IAB	Integrated Access Backhaul	OFDM	Orthogonal Frequency Division Multiplexing
BN	Batch Normalization	IDS	Intrusion Detection System	OSA	Opportunistic Spectrum Access
BS	Base Station	IIoT	Industrial Internet of Things	PDMA	Pattern Division Multiple Access
BW	Bandwidth	IoE	Internet of Everything	PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
C-RAN	Cloud Radio Access Network	IoT	Internet of Things	PRT	Perception-Reaction Time
CF	Cell Free	IoV	Internet of Vehicles	PSM	Problem Solving Methods
CNN	Convolutional Neural Network	IRS	Intelligent Reflecting Surface	PSO	Particle Swarm Optimization
CPS	Cyber Physical System	IWN	Industrial Wireless Networks	QoE	Quality of Experience
CR	Cognitive Radio	k-NN	k-Nearest Neighbors	QoS	Quality of Service
CSA	Cell-to-Switch Assignment	KPI	Key Performance Indicator	QPSK	Quadrature Phase Shift Keying
CPU	Central Processing Unit	LAA	Licensed-Assisted Access	R2	R-squared
CSI	Channel State Information	LACA	Learning-Assisted Clustered Access	RAN	Radio Access Network
D2D	Device to Device	LB	Load Balancing	RAT	Radio Access Technology
DBN	Deep Belief Network	LDR	Load-input Data Relationship	RF	Random Forest
DL	Deep Learning	LogReg	Logistic Regression	RIS	Reconfigurable Intelligent Surfaces
DNN	Deep Neural Networks	LS	Least Square	RL	Reinforcement Learning
DQL	Deep Q-Learning	LSTM	Long Short Term Memory	RLS	Recursive Least Square
DQN	Deep Q Network	LTE	Long Term Evolution	RNE	Remaining Node Energy
DRL	Deep Reinforcement Learning	MAB	Multi-Armed Bandit	ROC	Receiver Operating Characteristics
ECS	Edge Computing Server	MAPE	Mean Absolute Percentage Error	RRH	Remote Radio Head
ED	Edge Device	MARL	Multi-Agent Reinforcement Learning	RSRP	Reference Signal Received Power
EDLB	Effective Dynamic Load Balancing	MCC	Mathews Correlation Coefficient	RNN	Recurrent Neural Network
EE	Energy Efficiency	MC-NOMA	Multi-Carrier Non-Orthogonal Multiple Access	RMSE	Root Mean Squared Error
EM	Expectation Maximization	MDP	Markov Decision Process	SBS	Small cell Base Station
ETSI	European Telecommunications Standards Institute	MEC	Multi-access Edge Computing	SC	Small Cell
E2E	End-to-End	MEC-ISG	MEC-Industry Specification Group	SDN	Software Defined Networking
F-RAN	Fog Heterogeneous Radio Access Network	MIMO	Multiple Input Multiple Output	SE	Spectral Efficiency
FEC	Forwarding Error Correction	mIoT	massive Internet of Things	SINR	Signal to Interference Noise Ratio
FFDNet	Flexible Denoising Convolutional Neural Network	MM	Mobility Management	SISO	Single Input Single Output
SLA	Service-Level Agreement	UDN	Ultra Dense Network	VFC	Vehicular Fog Computing
SL	Supervised Learning	UDSC	Ultra-dense Small Cells	VLC	Visible Light Communication
SON	Self Organized Network	UE	User Equipment	VNF	Virtual Network Functions
SRS	Sounding impacting Reference Signals	UHD	Ultra High Definition	VR	Virtual Reality
SU	Secondary User	UL	Unsupervised Learning	VSN	Vehicular Social Networks
SVM	Support Vector Machine	URL	Uniform Resource Locator	VUE	Vehicle User Equipment
SWIPT	Simultaneous Wireless Information and Power Transfer	URLLC	Ultra-Reliable Low-Latency Communication	vRANs	virtualized Radio Access Networks
TPU	Tensor Processing Unit	USTLD	Uncoded Space-Time Labeling Diversity	WLAN	Wireless Local Area Network
TSC	Time Specific Communication	UAV	Unmanned Aerial Vehicles	XR	Extended Reality
VeFN	Vehicular Fog Network	MMSE	Minimum Mean Squared Error	ZF	Zero-Forcing
mMIMO	massive-Multiple Input Multiple Output				

ABBREVIATIONS

The acronyms used in this manuscript are listed in Table 9.

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